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# Phylogeny and evolutionary history of the genus *Nectogale* (Soricidae) inferred from whole–genome analyses, with description of a new species endemic to the Himalayas

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ABSTRACT

The genus *Nectogale*, endemic to the Himalayas, Hengduan Mountains, and adjacent ranges, comprises highly specialized semi-aquatic shrews adapted to high-elevation streams. Its diversity has likely been underestimated because of limited sampling and morphological conservatism, leaving species limits and evolutionary relationships unresolved. Here, we combined whole–genome data with morphological evidence to delimit species, infer phylogenetic history, and test for introgression and demographic change. Our results support three species: *Nectogale elegans*, *Nectogale sikhimensis*, and a newly identified species, *Nectogale jilongensis* sp. nov. Divergence-time analyses indicated that the *Nectogale* lineage diverged from its sister genus in the Middle Miocene (~14 Ma), followed by diversification between the Late Miocene (~7 Ma) and Early Pliocene (~4.5 Ma), broadly coinciding with Himalayan uplift and regional climatic transitions. Admixture analyses and D-statistics (ABBA–BABA tests) indicated significant introgression near species range boundaries, while pairwise sequentially Markovian coalescent (PSMC) analyses revealed contrasting population histories among the three lineages. These results highlight the role of contact zones in facilitating lineage divergence through gene flow, and suggest that regional environmental heterogeneity contributes to distinct evolutionary trajectories across the Himalayas. Our study revises *Nectogale* taxonomy and clarifies its evolutionary history, offering new insights into diversification in the Himalayan biodiversity hotspot.

## 1. Introduction

The Himalayas are among the world's most prominent biodiversity hotspots, and their exceptional diversity is often linked to complex geological and climatic processes associated with Cenozoic uplift of the Qinghai-Xizang Plateau (Ding et al., 2020; Lai et al., 2025; Xu et al., 2021; Xing and Ree, 2017). The region's steep elevational gradients, pronounced climatic heterogeneity, and multiple geographic barriers likely interact to promote habitat fragmentation, niche differentiation, and population isolation, creating conditions conducive to lineage divergence and the accumulation of endemic diversity (Lai et al., 2025; Lei et al., 2015; Manish and Pandit, 2018; Pandit et al., 2025). Although the Himalayan region has long attracted the attention of evolutionary biologists and has been studied across diverse organismal groups,

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including plants, birds, reptiles, and mammals, the region's biodiversity and the processes that generated it remain only partly resolved (Sharma and Hope, 2026).

In recent years, large-scale expeditions such as the Second Qinghai-Xizang Plateau Scientific Expedition and Research have continued to expand and revise regional species inventories (Chen et al., 2024a; Chen et al., 2021, 2024b; Pei et al., 2024; Wang et al., 2024; Zhu et al., 2025). Among small mammals alone, several previously overlooked Himalayan taxa have recently been described, including a new genus and species of scalopine mole, *Alpiscaptulus medogensis* (Chen et al., 2021), and a new genus and species of birch mouse, *Breviforamen shannanensis* (Zhu et al., 2025); three new species of Himalayan shrews in *Soriculus* (Chen et al., 2024b; Jiang et al., 2023; Pei et al., 2024); and 10 new species of mountain voles in *Neodon* (Liu et al., 2022; Wang et al., 2024; Wang et al., 2025), among others. Nevertheless, rugged terrain and limited accessibility can obscure true biodiversity, while morphological conservatism can hinder reconstruction of evolutionary history, particularly the timing and mechanisms of lineage divergence, the extent of gene flow, and the formation and maintenance of species boundaries. Systematic field surveys combined with population genomic analyses are therefore crucial for revealing hidden diversity and resolving the evolutionary processes that have shaped Himalayan biota.

The elegant water shrews of the genus *Nectogale* Milne-Edwards, 1870 (Mammalia, Eulipotyphla, Soricidae) are endemic to the Himalayan–Hengduan Mountains region and adjacent ranges, occurring in southwestern China, India, Nepal, Bhutan, and northern Myanmar (Wilson and Mittermeier, 2018). Species in this genus are generally restricted to high-elevation environments, inhabiting cold, fast-flowing streams within forested mountain landscapes, a highly specialized ecological niche among small mammals (Burgin and He, 2018). *Nectogale* is among the larger shrews in Soricidae (> 20 g) and is notable for its distinctive morphology and aquatic specializations, particularly webbed feet (Milne-Edwards, 1870), making it one of the most specialized semi-aquatic small mammals of mountain ecosystems. Despite these specializations, studies of the biology and evolution of *Nectogale* remain limited. Small population sizes, fragmented distributions, and confinement to high-elevation streams in remote mountain forests constrain field sampling and systematic surveys, leaving its diversity, evolutionary history, and population structure poorly understood.

The taxonomy of *Nectogale* has long been controversial. Milne-Edwards (1870) described the first species in the genus, *Nectogale elegans*, based on a specimen from Muping (Moupin) Town, Baoxing County, Sichuan Province, China. De Winton and Styan (1899) examined two *Nectogale* specimens from Sikkim and noted a browner overall pelage, a less distinct boundary between dorsal and ventral coloration, and a shorter-cusped first upper incisor; on this basis they described *N. sikhimensis*. The validity of *N. sikhimensis* was, however, inconsistently recognized thereafter. For example, Allen (1938), Ellerman and Morrison-Scott (1951), Zhang (1999), Wang (2003), and Smith and Xie (2020) treated the Sikkim web-footed shrew as a subspecies of *N. elegans* (*N. e. sikhimensis*), whereas Walker (1975), Feng et al. (1986), Hoffmann (1987), Hutterer, 2005b), and Wilson and Mittermeier (2018) regarded it as a synonym of *N. elegans*. Although Fan et al. (2022) recently revalidated *N. sikhimensis* as a distinct species by integrating morphological and multilocus genetic data, their conclusions were based on relatively few specimens and incomplete geographic sampling. Such limitations can reduce confidence in inferences about species boundaries and intrageneric variation. As a result, key questions about the true diversity, geographic structure, and evolutionary processes within *Nectogale* remain unresolved, particularly with respect to historical isolation, the delimitation of species boundaries, and potential gene flow among lineages.

Against this background, the alpine, semi-aquatic genus *Nectogale* provides an informative system for investigating species divergence, gene flow, and the formation of species boundaries in the Himalayas. Between 2017 and 2024, we conducted systematic field surveys in the Himalayas and southwestern China and collected specimens spanning the known and putative distributional range of *Nectogale*. Preliminary analyses revealed that nine specimens from Jilong and Dingjie were deeply divergent from the two recognized species in cytochrome *b* (Cyt b) sequences. These specimens were also geographically distinct and showed clear morphological differences, particularly in pelage coloration and the shape of the coronoid process, rendering them unassignable to any known species. We therefore tentatively treated them as a putative new species, hereafter *Nectogale* sp. Using these new samples, we combined whole–genome resequencing with detailed morphological analyses to infer phylogenetic relationships, delimit species boundaries, evaluate population structure and introgression, and reconstruct demographic history. More broadly, we assessed whether diversification and lineage interactions within *Nectogale* are consistent with roles for geographic isolation and Himalayan geological and climatic events during speciation.

## 2. Material and Methods

### 2.1. Sampling

Our sampling covered most of the southern Hengduan Mountains and the Himalayas (Fig. 1; Table S1). We analysed whole–genome resequencing data for 21 *Nectogale* individuals and 11 outgroup samples (nine newly sequenced and two publicly available genomes; Table S2). To validate specimen identities, we integrated findings from Fan et al. (2022), diagnostic morphological features, geographic distributions, a newly reconstructed Cyt b gene tree (Fig. S1; Table S3), and Kimura 2-parameter (K2P) distances for Cyt b (Table 1). These samples represent three operational taxonomic units: two recognized species (*N. elegans* and *N. sikhimensis*) and one putative lineage (*Nectogale* sp.).

For morphological analyses, we examined 42 museum specimens covering the major geographic range of the genus (Table S1); only adults with fully erupted permanent molars ($M^1$–$M^3$/$M_1$–$M_3$) were included.

Muscle or liver tissue for sequencing was preserved in ethanol in the field and stored at − 80 °C. All tissues and voucher specimens are deposited at the Kunming Institute of Zoology, Chinese Academy of Sciences. Animal handling complied with the guidelines of the American Society of Mammalogists (Sikes, 2016). The protocols were approved by the internal review board of Kunming Institute of Zoology, with permission from relevant local authorities (Approval No. SMKX-20191020–212).

### 2.2. Genome sequencing and variant calling

Whole–genome resequencing was performed on the DNBSEQ-T7 platform (Table S2). Raw reads were quality-filtered and adapter-trimmed with fastp v0.23.4 (Chen, 2023).

To establish a reference for downstream mapping, we generated a de novo reference genome for a female *N. elegans* specimen (Field No. EG202311299) using MEGAHIT v1.2.9 (Li et al., 2015) and RagTag v2.1.0 (Alonge et al., 2022), and assembly completeness was assessed with Compleasm v0.2.6 (Huang and Li, 2023) against the mammalia_odb10 dataset from the OrthoDB database (Manni et al., 2021). Gene annotation was transferred with Liftoff v1.6.3 (Shumate and Salzberg, 2021), using *Sorex araneus* (RefSeq accession: GCF_027595985.1) as the reference, and coding and protein sequences were extracted with GffRead v0.12.7 (Pertea and Pertea, 2020).

Clean reads were mapped to the *N. elegans* reference genome using BWA-MEM v0.7.19 (Li, 2013). PCR duplicates were marked and removed using Picard v3.3.0 (Broad Institute, https://broadinstitute.github.io/picard/). For population genetic analysis, we called high-quality biallelic SNPs and genotypes using BCFtools v1.19 (Danecek et al., 2021). We retained SNPs with a missing data rate of < 20% using VCFtools v0.1.16 (Danecek et al., 2011). Detailed workflows, assembly parameters, annotation procedures, and filtering thresholds are

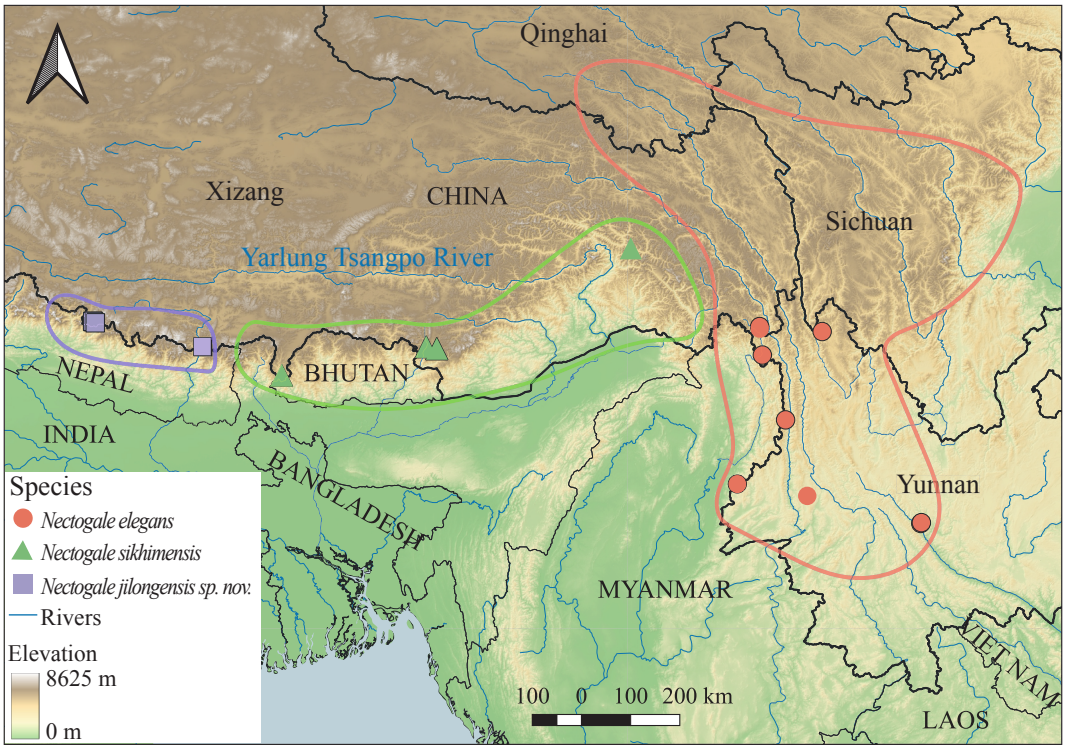


**Fig. 1.** Geographic range of the genus *Nectogale*. Symbols indicate the sampling localities of the resequenced individuals (Table S1); coloured outlines indicate the known distributions of the three species recognized in this study, based on our records and published localities (Table S3).

**Table 1**
Genetic divergence among *Nectogale* species.

| Species | $\pi$ | $F_{ST}$ | $D_{XY}$ | *K2P–Cyt b* |
|---|---|---|---|---|
| *N. elegans* | 0.007 | 0.816a | 0.057a | 0.139a |
| *N. sikhimensis* | 0.012 | 0.815b | 0.058b | 0.129b |
| *Nectogale* sp. | 0.011 | 0.558c | 0.030c | 0.077c |

a. *N. elegans* vs *N. sikhimensis*, b. *N. elegans* vs *Nectogale* sp., c. *N. sikhimensis* vs *Nectogale* sp.

provided in the Supplementary Methods.

### 2.3. Phylogenetic analyses and divergence times

First, we reconstructed phylogenies from the mitochondrial and nuclear genomes separately using a concatenation approach, because the two genomes are inherited differently (maternally versus biparentally) and may reflect different evolutionary histories. The nuclear genome served as our primary evidence, with the mitochondrial genome as an independent maternal marker corroborating lineage monophyly.

For the mitochondrial dataset, mitochondrial genomes were assembled from 5 Gb of clean reads per sample using GetOrganelle v1.7.7.1 (Jin et al., 2020) and annotated with MitoZ v3.6 (Meng et al., 2019). We extracted the 13 protein-coding genes and two ribosomal RNA genes from each mitogenome, aligned each gene with MAFFT v7.525 (Katoh and Standley, 2013), and concatenated the 15 gene regions using the "Concatenate Sequence" function in PhyloSuite v1.2.3 (Zhang et al., 2020) to construct a mitochondrial supermatrix of 14,075 bp.

For the nuclear dataset, consensus sequences for each resequenced sample were generated with BCFtools v1.19. We performed homology inference for all species using OrthoFinder v2.5.5 (Emms and Kelly, 2019), and identified single-copy genes. Genes whose coding sequences were not divisible by three or contained internal stop codons were excluded. The remaining loci were aligned with MAFFT (Katoh and Standley, 2013) and trimmed with trimAl v1.5.rev0 (Capella-Gutiérrez et al., 2009). These trimmed loci were then concatenated using the "Concatenate Sequence" function in PhyloSuite to construct a nuclear genome supermatrix of 9,482,481 bp.

Maximum-likelihood (ML) trees were then reconstructed for the nuclear and mitochondrial supermatrices separately in IQ-TREE v2.4.0 (Minh et al., 2020), with automated model selection by ModelFinder (+MERGE) (Kalyaanamoorthy et al., 2017) and branch support from 10,000 ultrafast bootstrap (UFBoot) replicates (Hoang et al., 2018) and 1,000 SH-aLRT replicates (Guindon et al., 2010).

Second, we inferred a species tree under the coalescent with CASTER v1.20.2.5 (Zhang et al., 2025) using the 5,816 trimmed single-copy gene alignments, with individuals assigned to their respective species.

Finally, divergence times were estimated with MCMCTree in PAML v4.10.7 (Yang, 2007), using the CASTER species tree as the fixed topology. For computational efficiency, we used four-fold degenerate (4DTV) sites, which are expected to be least affected by selection at the protein level, extracted from the concatenated nuclear supermatrix. We conducted two independent MCMC runs with different random seeds; each run consisted of 100 million iterations, discarding the first 10 million iterations (10%) as burn-in. Samples were then drawn every 10 iterations, yielding 9 million samples per run. We applied three secondary calibrations from the broader soricid framework of Yuan et al. (2024), which is based on a more comprehensive sampling of Soricidae fossils: (1) Soricidae, 51.06 Ma (95% CI: 41.30–64.34 Ma); (2) Nectogalini, 20.52 Ma (95% CI: 14.87–27.70 Ma); and (3) the split of *Chimarrogale–Nectogale*, 13.37 Ma (95% CI: 8.31–19.65 Ma). Convergence was assessed in Tracer v1.7.2 (Rambaut et al., 2018), confirming effective sample sizes (ESS) $>$ 200 for all parameters and consistent posterior estimates between runs.

### 2.4. Species delimitation

To quantify genetic differentiation among the three *Nectogale* lineages, we calculated K2P distances for Cyt b in MEGA X (Kumar et al., 2018), because this marker is widely used for species-level

comparisons in mammals (Bradley and Baker, 2001). Using the all-sites dataset (46,875,923 biallelic SNPs and 349,050,675 invariant sites), we estimated nucleotide diversity ($\pi$) within each lineage, the pairwise fixation index (FST), and absolute divergence (DXY; the mean number of pairwise nucleotide differences per site between lineages) with pixy 2.0.0.beta8 (Korunes and Samuk, 2021), in non-overlapping 50-kb windows across the genome. To reduce potential artifacts from poorly assembled regions, we excluded windows located on scaffolds shorter than 1 Mb.

We also performed multispecies coalescent (MSC) species delimitation in BPP v4.8.4 (Yang, 2015) to test whether the three predefined lineages were supported as distinct species-level units. To reduce the risk of false positives (over-splitting) associated with an incorrect guide tree, we ran both the A10 model (species delimitation using a fixed guide tree) (Yang and Rannala, 2010) and the A11 model (joint species delimitation and species tree estimation) (Yang and Rannala, 2014), with priors $\theta \sim$ Gamma (2, 2000) and $\tau \sim$ Gamma (2, 1000), using 12 independent runs per model with different algorithm settings (Table S4; burn-in = 100,000 iterations; 100,000 samples; sampling interval = 10). For computational tractability, we randomly selected 500 loci from the single-copy gene set, a number of loci previously shown to be adequate for species delimitation (Tonzo et al., 2019).

### 2.5. Population genomic analyses

We assessed population structure using ADMIXTURE v1.3.0 (Alexander et al., 2009) based on SNP data. We evaluated values of K ranging from 1 to 6, with 20 independent runs for each value. Here, K represents the assumed number of ancestral populations. The best-supported value of K was identified using ADMIXTURE's cross-validation (CV) procedure. Replicate clustering solutions were summarized and visualized in pong v1.5 (Behr et al., 2016) using the greedy algorithm. We further summarized population structure using principal component analysis (PCA) in PLINK v1.9.0-b.7.7 (Purcell et al., 2007). Prior to PCA and admixture analyses, SNPs were pruned for linkage disequilibrium in PLINK using a sliding window of 50 SNPs, a step size of 10 SNPs, and a pairwise linkage disequilibrium ($r^2$) threshold of 0.1, yielding 24,274 unlinked biallelic SNPs.

To quantify introgression among *N. elegans*, *N. sikhimensis*, and *Nectogale* sp., we performed ABBA–BABA tests using Dsuite v0.5-r58 (Malinsky et al., 2021). Based on phylogenetic relationships and geographic distributions (Table S1), we grouped individuals into seven population-level units: two *N. elegans* populations (*N. elegans* 1 from western Yunnan; *N. elegans* 2 from central Yunnan), three *N. sikhimensis* populations (*N. sikhimensis* 1 from Bomi; *N. sikhimensis* 2 from Cuona; *N. sikhimensis* 3 from Yadong), and two *Nectogale* sp. populations (*Nectogale* sp. 1 from Jilong; *Nectogale* sp. 2 from Dingjie), with *Chimarrogale styani* as the outgroup. Because the D-statistic and $f_4$-ratio remain informative even when a population is represented by a single individual in genome-wide data, tests were conducted at the level of these population units.

We calculated D-statistics and $f_4$-ratios for all population trios with Dsuite Dtrios, estimated introgression along specific branches with the f-branch ($f_b$) statistic (Dsuite Fbranch), and visualized the results with dtools.py (https://github.com/millanek/Dsuite). We considered $|Z| > 3.09$ ($p < 0.001$) as evidence of significant introgression (Malinsky et al., 2021; Ding et al., 2025).

To infer demographic histories for the three *Nectogale* lineages, we used the pairwise sequentially Markovian coalescent (PSMC) model (Li and Durbin, 2011), selecting one high-coverage genome from each lineage (*N. elegans*, *N. sikhimensis*, and *Nectogale* sp.). For each selected genome, consensus sequences were generated from duplicate-marked BAM files using BCFtools and vcfutils, retaining sites with read depths between 10 and 100. We then converted the consensus sequences into PSMC input files and inferred changes in effective population size through time. We used a mutation rate of $2.54 \times 10^{-9}$ substitutions per site per generation, calculated following Li et al. (2025), and assumed a generation time of 1.04 years (Pacifici et al., 2013). The PSMC analysis was repeated with 100 bootstrap replicates to assess variation in the inferred trajectories.

### 2.6. Morphological analyses

Four external measurements, body mass (W), head–and–body length (HBL), hind foot length (HL), and tail length (TL) were recorded in the field. Body mass was measured to the nearest 0.1 g, whereas linear measurements were recorded to the nearest 0.1 mm. External measurements were used for the taxonomic descriptions but were excluded from statistical analyses to avoid inter-observer variation (Chen et al., 2017). In addition, we measured 22 craniodental variables using a digital caliper to 0.01 mm (Supplementary Methods; Fig. S2). We computed descriptive statistics (mean ± SD) and used one-way analysis of variance (ANOVA) to test for differences in each craniodental measurement among the three lineages. When ANOVA was significant ($p < 0.05$), Tukey's honestly significant difference (HSD) post hoc tests were used to identify pairwise differences. Terminology for morphological, cranial, and dental descriptions follows Hutterer , 2005a, Pan et al. (2007), and Chen et al. (2024b).

We used principal component analysis (PCA) and linear discriminant analysis (LDA) to summarize multivariate skull variation among *Nectogale* lineages. PCA was implemented with the *principal* function in the R package *psych* v2.4.6 (Revelle, 2026), after $\log_{10}$-transforming all craniodental variables to normalize the data and reduce scale differences among measurements. We also performed LDA in the R package *MASS* (Venables and Ripley, 2002) to identify the combination of measurements that best discriminated the three lineages. Variables were ranked by the absolute values of their discriminant coefficients in a full model and then added stepwise; classification accuracy at each step was assessed by leave-one-out cross-validation (LOOCV) to identify an optimal subset of variables.

## 3. Results

### 3.1. Molecular data and phylogenetic analyses

The *N. elegans* assembly was 2.25 Gb in length, with 80.36% BUSCO completeness (Fig. S3). In total, 3.93 Tb of clean data were obtained for the 30 newly sequenced samples (mean 130.95 Gb per sample; Table S2). SNP calling yielded 47,816,720 raw variants and 387,231,220 invariant sites, from which 46,875,923 high-quality biallelic SNPs and 349,050,675 invariant sites were retained after quality filtering. OrthoFinder identified 6,270 single copy orthologs shared by all samples, of which 5,816 passed the quality filters and were retained.

Both maximum-likelihood (ML) analyses of the concatenated nuclear and mitochondrial datasets strongly supported the monophyly of *Nectogale* (UFBoot = 100; SH-aLRT = 100; Fig. 2A, B). All nodes separating the focal lineages received maximal support, resolving three distinct clades corresponding to *N. elegans*, *N. sikhimensis*, and *Nectogale* sp. The two Xizang lineages (*N. sikhimensis* and *Nectogale* sp.) were recovered as sisters, and together they were sister to *N. elegans*. The coalescent species tree inferred from 5,816 single-copy genes supported the same topology (Fig. 2C). Topological discordance among the trees was confined to the outgroups, likely reflecting incomplete lineage sorting and mitonuclear discordance. Geographically, *N. elegans* occurs across the Hengduan Mountains (western Yunnan, Sichuan and southern Qinghai), *N. sikhimensis* occurs in southeastern and southern Xizang (Bomi to Yadong) and adjacent regions, and *Nectogale* sp. occurs in southern Xizang (Dingjie and Jilong).

Divergence-time estimates indicated that the split between *Nectogale* and *Chimarrogale* occurred at approximately 13.98 Ma (95% HPD = 11.03–16.89 Ma; Fig. 2D). The divergence between *N. elegans* and the (*N. sikhimensis* + *Nectogale* sp.) clade was estimated at 7.08 Ma

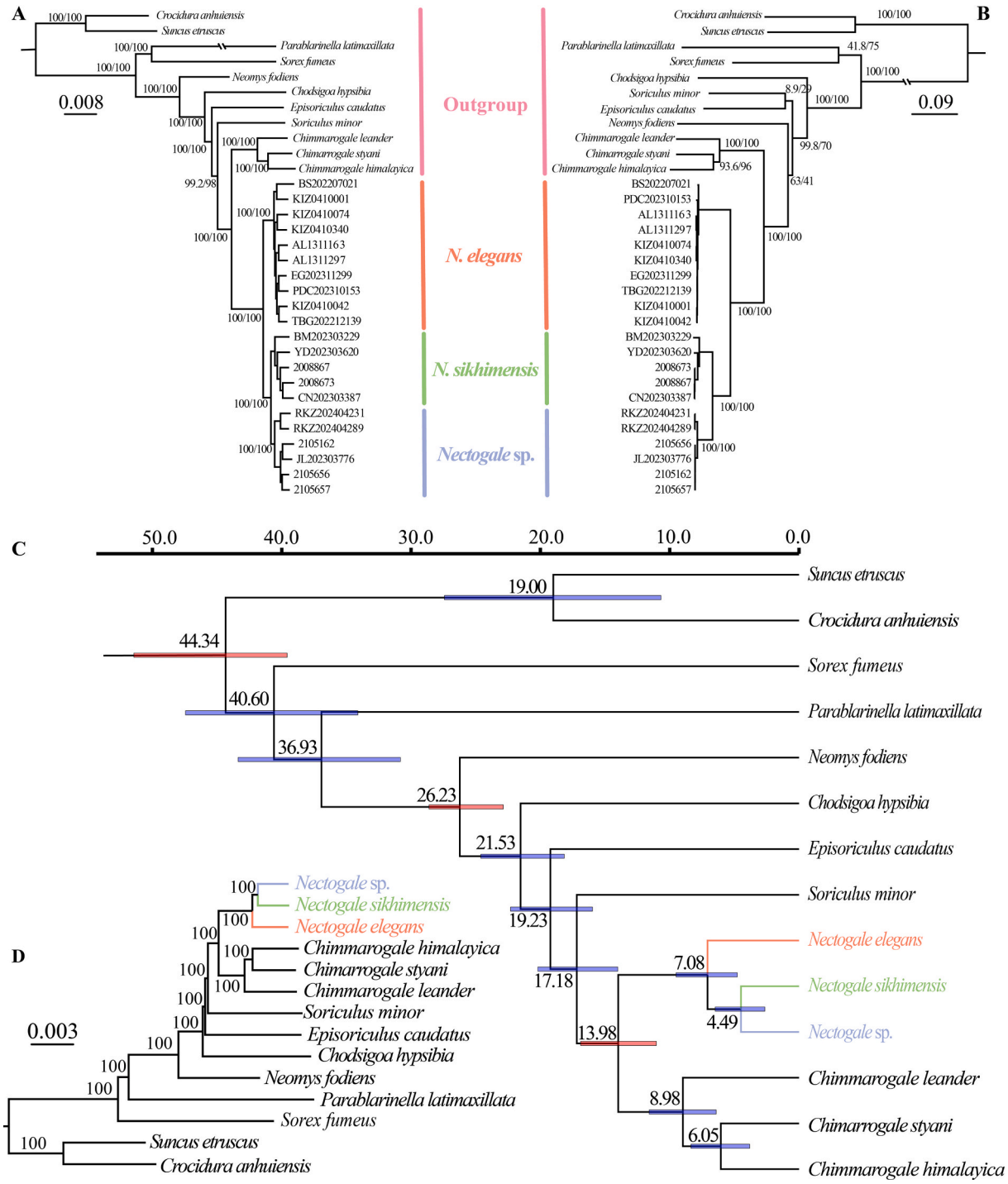


**Fig. 2.** Phylogenetic relationships and divergence times of *Nectogale*. (A) Maximum-likelihood (ML) tree based on the concatenated alignment of 5,816 nuclear single-copy genes (9,482,481 bp). (B) ML tree based on the 15 concatenated mitochondrial genes (14,075 bp). Node labels show SH-aLRT support (left) and ultrafast bootstrap support (UFBoot; right); scale bars are in substitutions per site. (C) Time-calibrated phylogeny inferred from 4DTV sites with MCMCTree in PAML v4.10.7. The x-axis shows time before present (Ma); node numbers are posterior median ages (Ma). Horizontal bars indicate 95% highest posterior density (HPD) intervals of node ages; red bars mark the three calibrated nodes (Section 2.3). (D) Coalescent species tree inferred from the 5,816 single-copy orthologs with CASTER v1.20.2.5; numbers at nodes are local support values, and the scale bar indicates branch length in CASTER units derived from site-pattern statistics. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(95% HPD = 4.77–9.51 Ma), and the split between *N. sikhimensis* and *Nectogale* sp. at 4.49 Ma (95% HPD = 2.64–6.48 Ma).

### 3.2. Species tree and species delimitation

Pairwise K2P distances for Cyt b ranged from 7.7% to 13.9% (Table 1). Genome-wide summary statistics also indicated strong differentiation, with pairwise $F_{ST}$ of 0.558–0.816 and $D_{XY}$ of 0.030–0.058 between lineages, and nucleotide diversity ($\pi$) of 0.007–0.012 within lineages (Table 1). BPP supported the three lineages as distinct species (posterior probability, PP = 1.0) under both the A10 and A11 models and all algorithm settings tested (Table S4).

### 3.3. Population structure and demography

Principal component analysis based on the 24,274 LD-pruned SNPs clearly separated the three lineages, with PC1 and PC2 explaining 43.93% and 17.90% of the variance, respectively (Fig. 3A). Although the cross-validation error was lowest at K = 2, it was only slightly higher at K = 3 (Fig. 3B), and the K = 3 solution (recovered in 17 of 20 runs) was concordant with the PCA clusters, the species-tree topology, and the BPP results; we therefore consider K = 3 the most biologically meaningful solution. Accordingly, the three ancestry components were interpreted as corresponding to the three lineages (Fig. 3C). One *N. sikhimensis* individual from Bomi and two *Nectogale* sp. individuals from Dingjie showed partial admixture with neighboring lineages (Fig. 3A, C).

The ABBA–BABA tests identified significant introgression among all three lineage pairs (Table S5): *N. elegans*–*N. sikhimensis* ($D$ = 0.020, $f_4$-ratio = 0.008), *N. elegans*–*Nectogale* sp. ($D$ = 0.021, $f_4$-ratio = 0.018), and *N. sikhimensis*–*Nectogale* sp. ($D$ = 0.039, $f_4$-ratio = 0.037), where non-zero $D$ values indicate asymmetric allele sharing and the $f_4$-ratio estimates the proportion of introgressed ancestry. Introgression signals were strongest between geographically adjacent units: western Yunnan population of *N. elegans* (*N. elegans* 1) showed significant introgression with both *N. sikhimensis* and *Nectogale* sp., and the Dingjie population of *Nectogale* sp. (*Nectogale* sp. 2) showed strong introgression with *N. sikhimensis* from Yadong (*N. sikhimensis* 3) (Fig. 3D).

Demographic reconstructions revealed distinct trajectories among the three lineages, but all three showed pronounced declines in inferred effective population size ($N_e$) toward the present (Fig. 3E). *N. elegans* experienced a sustained decline from the Naynayxungla glaciation (NG; ~500–780 kya) through the Penultimate Glaciation (PG; ~135–194 kya), followed by a modest recovery and a subsequent decline. *N. sikhimensis* expanded during the NG, declined thereafter, recovered briefly after the PG (peaking at ~ 60–70 kya), and then declined continuously. *Nectogale* sp., whose trajectory could be reconstructed only back to ~ 400 kya (see Section 4.2), showed a modest expansion before the PG followed by a sustained decline.

### 3.4. Morphological variation among Nectogale lineages

The three lineages differed in body size and craniodental morphology (Table 2). *Nectogale* sp. was the largest (mean CIL = 26.02 mm, W = 43.35 g), followed by *N. elegans* (CIL = 25.45 mm, W = 39.68 g) and *N. sikhimensis* (CIL = 24.91 mm, W = 34.27 g). One-way ANOVA identified significant differences among lineages for 15 of the 22 craniodental variables (Table 2); postpalatal depth (PPD), mandibular length (MnL), and length of the lower molar row (LLM) differed significantly in all three pairwise comparisons (Tukey's HSD; Fig. S4).

Multivariate analyses also separated the three lineages. In the PCA of

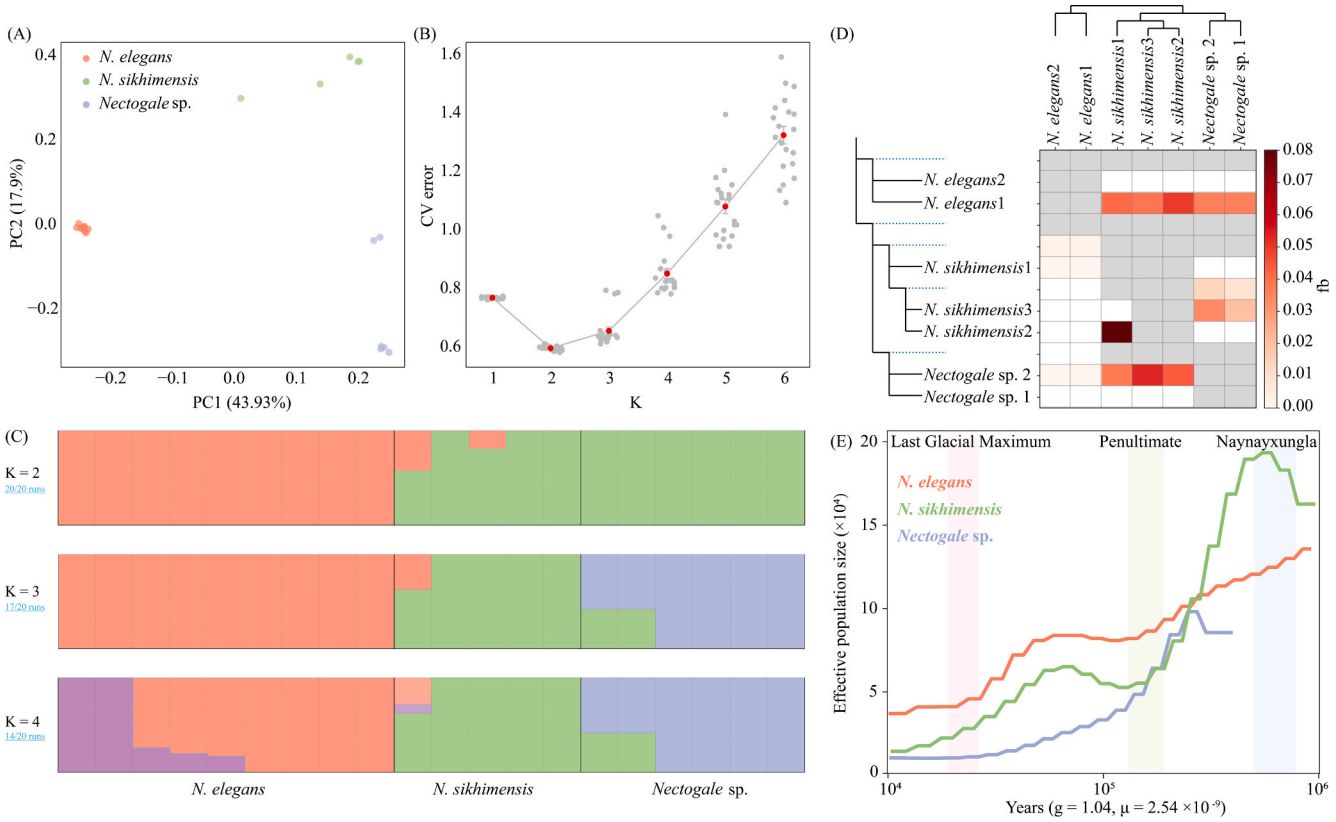


**Fig. 3.** Genome-wide population structure, gene flow, and demographic history of *Nectogale*. (A) Principal component analysis (PCA) based on 24,274 LD-pruned biallelic SNPs; the scatterplot shows the first two principal components (PC1 and PC2), with the percentage of variance explained on each axis, for *N. elegans* (orange), *N. sikhimensis* (green), and *Nectogale* sp. (blue). (B) ADMIXTURE cross-validation (CV) error for K = 1–6, where K is the assumed number of ancestral populations; grey points show the 20 independent runs per K and red points their mean. (C) ADMIXTURE ancestry proportions for K = 2, 3 and 4 (rows), showing the major mode across the 20 runs (number of runs supporting each mode indicated at left). Each vertical bar represents one individual; colours indicate inferred ancestry components. (D) f-branch statistic ($f_b$) among the seven population units (defined in Section 2.5 and Table S5) estimated with Dsuite; grey cells denote combinations that cannot be tested. (E) Demographic histories inferred with the pairwise sequentially Markovian coalescent (PSMC) for one genome per lineage. The x-axis (log scale) shows time in years, scaled with a generation time (g) of 1.04 years and a mutation rate (μ) of 2.54 × $10^{-9}$ per site per generation; the y-axis shows effective population size (×$10^4$). Shaded bands mark the Last Glacial Maximum, the Penultimate Glaciation and the Naynayxungla Glaciation. The trajectory of *Nectogale* sp. spans a shorter period because PSMC scales time by the genome-specific $N_0$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

**Table 2**
Summary statistics (mean ± SD, range, and sample size) for external and craniodental measurements of *Nectogale* specimens. Body mass (W) is reported in grams, and all linear measurements are reported in millimeters. Measurement abbreviations are defined in Supplementary Methods.

| Variable | *N. elegans* (N = 25) | *N. sikhimensis* (N = 11) | *Nectogale* sp. (N = 6) | *F–statistic* |
|---|---|---|---|---|
| W | 39.68 ± 9.53<br>27.50–60.70<br>(22) | 34.27 ± 5.97<br>24.90–43.10(11) | 43.35 ± 3.73<br>38.10–48.80(6) | |
| HBL | 101.39 ± 9.64<br>84.00–120.00<br>(23) | 98.55 ± 8.63<br>90.00–120.00<br>(11) | 105.50 ± 6.75<br>95.00–115.00<br>(6) | |
| TL | 107.26 ± 7.40<br>92.00–121.00<br>(23) | 108.36 ± 3.11<br>105.00–113.00<br>(11) | 115.67 ± 2.73<br>112.00–120.00<br>(6) | |
| HL | 26.68 ± 2.13<br>22.00–30.00<br>(22) | 26.82 ± 1.66<br>25.00–31.00(11) | 27.00 ± 3.58<br>20.00–30.00(6) | |
| CIL | 25.45 ± 0.63<br>24.56–26.68<br>(25) | 24.91 ± 0.88<br>23.74–26.16 (11) | 26.02 ± 0.38<br>25.4–26.34 (6) | 5.53** |
| PIL | 12.49 ± 0.34<br>11.64–13.1 (25) | 11.92 ± 0.56<br>11.13–12.78 (11) | 12.52 ± 0.31<br>11.96–12.82 (6) | 8.19** |
| PPL | 9.21 ± 0.36<br>8.72–10.07 (25) | 9.16 ± 0.25<br>8.81–9.63 (11) | 9.41 ± 0.15<br>9.22–9.57 (6) | 1.28 |
| UTL | 11.59 ± 0.28<br>11.09–12.07<br>(25) | 11.11 ± 0.49<br>10.4–11.63 (11) | 11.78 ± 0.35<br>11.1–12.14 (6) | 9.45*** |
| $P^4M^3$ | 6.32 ± 0.17<br>6.05–6.67 (25) | 6.09 ± 0.21<br>5.73–6.44 (11) | 6.36 ± 0.1<br>6.24–6.47 (6) | 7.39** |
| IOB | 6.33 ± 0.23<br>6.01–6.83 (25) | 6.1 ± 0.18<br>5.79–6.36 (11) | 6.42 ± 0.08<br>6.32–6.53 (6) | 6.85** |
| CB | 15.34 ± 0.51<br>14.73–16.58<br>(25) | 15.07 ± 0.43<br>14.54–15.99 (11) | 15.49 ± 0.3<br>15.21–15.87 (6) | 1.93 |
| CH | 8.65 ± 0.18<br>8.37–8.99 (25) | 8.56 ± 0.3<br>8.17–9.11 (11) | 8.71 ± 0.23<br>8.3–8.89 (6) | 1.08 |
| $M^2M^2$ | 7.9 ± 0.28<br>7.42–8.48 (25) | 7.55 ± 0.23<br>7.11–7.88 (11) | 7.99 ± 0.26<br>7.6–8.22 (6) | 8.12** |
| BMF | 5.01 ± 0.2<br>4.75–5.38 (25) | 5.1 ± 0.17<br>4.85–5.41 (11) | 5.23 ± 0.1<br>5.1–5.38 (6) | 3.77* |
| RL | 9.93 ± 0.45<br>8.79–10.94 (25) | 9.31 ± 0.31<br>8.81–9.65 (11) | 10.26 ± 0.36<br>9.66–10.66 (6) | 13.42*** |
| PRL | 14.66 ± 0.45<br>13.78–15.46<br>(25) | 14.44 ± 0.52<br>13.58–15.3 (11) | 14.7 ± 0.33<br>14.31–15.05 (6) | 0.98 |
| ZP | 2.69 ± 0.12<br>2.35–2.91 (25) | 2.41 ± 0.18<br>2.16–2.76 (11) | 2.76 ± 0.29<br>2.24–2.99 (6) | 12.55*** |
| MB | 9.17 ± 0.36<br>8.71–9.98 (25) | 8.67 ± 0.3<br>8.18–9.1 (11) | 9.28 ± 0.05<br>9.21–9.36 (6) | 10.76*** |
| PPD | 5.16 ± 0.18<br>4.84–5.5 (25) | 4.8 ± 0.12<br>4.47–4.94 (11) | 4.9 ± 0.03<br>4.87–4.95 (6) | 21.82*** |
| HCP | 6.03 ± 0.23<br>5.64–6.57 (25) | 5.81 ± 0.24<br>5.5–6.2 (11) | 6.11 ± 0.05<br>6.04–6.17 (5) | 4.84* |
| HCV | 3.94 ± 0.19<br>3.54–4.3 (24) | 3.82 ± 0.16<br>3.62–4.11 (11) | 3.9 ± 0.08<br>3.81–4.01 (5) | 1.78 |
| HAC | 4.93 ± 0.24<br>4.4–5.39 (25) | 4.84 ± 0.24<br>4.47–5.22 (11) | 5.02 ± 0.08<br>4.9–5.1 (5) | 1.2 |
| LLI | 4.94 ± 0.41<br>3.62–5.48 (25) | 5.2 ± 0.46<br>4.44–5.81 (11) | 5.13 ± 0.3<br>4.72–5.51 (5) | 1.77 |
| LLM | 5.3 ± 0.12<br>5.06–5.55 (25) | 5.18 ± 0.13<br>4.92–5.4 (11) | 5.38 ± 0.06<br>5.29–5.43 (5) | 6.17** |
| LTR | 7.03 ± 0.15<br>6.71–7.26 (25) | 6.85 ± 0.2<br>6.54–7.1 (11) | 7.08 ± 0.07<br>6.98–7.14 (5) | 5.57** |
| MnL | 11.9 ± 0.36<br>11.36–12.72<br>(25) | 11.32 ± 0.4<br>10.55–11.9 (11) | 12.29 ± 0.17<br>12.07–12.53 (5) | 15.47*** |

*$P < 0.05$; **$P < 0.01$; ***$P < 0.001$. External measurements (W, HBL, TL, HL) were not tested (blank F-statistic cells).

42 skulls, PC1 (61.48% of the variance) represented a size gradient along which *N. sikhimensis* occupied mainly negative scores while *N. elegans* and *Nectogale* sp. overlapped broadly; PC2 (7.79%) was loaded primarily by lower incisor length (LLI) (Fig. 4A; Table 3). LDA based on 12 craniodental variables achieved 97.6% classification accuracy under LOOCV. The first linear discriminants (LD1, 79.8% of the discriminant variance) loaded primarily on lower tooth-row length (LTR), palato-incisive length (PIL), and postpalatal length (PPL), whereas the second linear discriminants (LD2, 20.2%) loaded primarily on postpalatal depth (PPD), mandibular length (MnL), and length of the lower molar row (LLM) (Fig. 4B).

Qualitative examination revealed diagnostic differences in cranial architecture and pelage (Figs. 5, 6). *N. sikhimensis* had the smallest body size, with a distinctly higher and more inflated braincase, grayish-black pelage, and poorly developed caudal fimbriae with a brown-black tip. *N. elegans* had intermediate body size, with a flattened braincase, brownish-gray pelage, and the longest and best-developed caudal fimbriae, with many long brown fimbriae at the tail tip. *Nectogale* sp. had the largest body size, with a moderately flattened braincase, lighter gray pelage, and less-developed caudal fimbriae ending in a slender tip with bright white bristles.

The coronoid process differed markedly among lineages. *Nectogale* sp. had a wider and more robust coronoid process with a distinct oblique surface on the lateral side and an overall block-like form (Fig. 6C5). By contrast, *N. elegans* had a rounder, more slender coronoid process that tapered gradually to a pointed apex, giving it an arched, streamlined appearance (Fig. 6A5). The coronoid process of *N. sikhimensis* was intermediate between the two, with a dagger-like shape (Fig. 6B5). Upper unicuspid morphology provided additional diagnostic characters. In *Nectogale* sp., all upper unicuspids showed a distinctive specialization, with smaller but sharper cusps, twisted bases, and thin cingula (Fig. 6C6). In *N. sikhimensis* and *N. elegans*, the second upper unicuspid was sickle-shaped; the third upper unicuspid of *N. elegans* was similar to the second, whereas that of *N. sikhimensis* was more triangular in overall outline (Fig. 6A6, B6).

### 3.5. Systematics

*Nectogale jilongensis* sp. nov.

**Suggested common name:** Jilong Elegant Water Shrew; 吉隆蹼足鼩.

urn:lsid:zoobank.org:pub:CFA0439F-CBC5-440B-B14F-3504BEB53510

Type locality. Jilong County, Rikaze, Xizang, China

Type material

**Holotype.** JL202303775, adult female, collected on 24 May 2023 by Changzhe Pu at Jilong County, Rikaze, Xizang, China (28.363341°N, 85.361739°E; 2355 m a.s.l.).

Dried skin, cleaned skull, and muscle tissue are deposited at the Kunming Institute of Zoology (KIZ), Chinese Academy of Sciences.

**Paratypes.** Three adult females: JL202303780 and JL202303791, collected in Jilong County, Rikaze, Xizang, China, in May 2023 (2355 m a.s.l.), and RKZ202404289, collected in Dingjie County, Rikaze, Xizang, China, in April 2024 (3059 m a.s.l.). All are deposited at KIZ.

**Etymology.** The specific epithet *jilongensis* refers to Jilong, the type locality, with the Latin adjectival suffix *–ensis* ('belonging to').

**Diagnosis.** The largest species of *Nectogale*. Dorsal pelage gray from head to rump, interspersed with lustrous white guard hairs; overall lighter than *N. elegans* (brownish) and *N. sikhimensis* (darker, nearly black). Tail base dark gray with short white bristles; caudal fimbriae relatively underdeveloped and uniformly lustrous white (well developed, with brown terminal tufts, in *N. elegans*; poorly developed, with short, sparse brown terminal tufts, in *N. sikhimensis*). Upper unicuspids with small but sharp cusps, twisted bases, and thin, elongated cingula. Coronoid process wide and robust, columnar to nearly quadrangular, bearing a distinct oblique facet on the lateral surface (gradually tapering and arcuate in *N. elegans* and *N. sikhimensis*). Known only from southern Xizang, between Dingjie and Jilong, west of Mount Kangchenjunga.

**Description.** *Nectogale jilongensis* sp. nov. is the largest species in the genus, with a body mass of 43.35 ± 3.73 g and a head–and–body length of 105.50 ± 6.75 mm (Table 2). The dorsal pelage is gray with

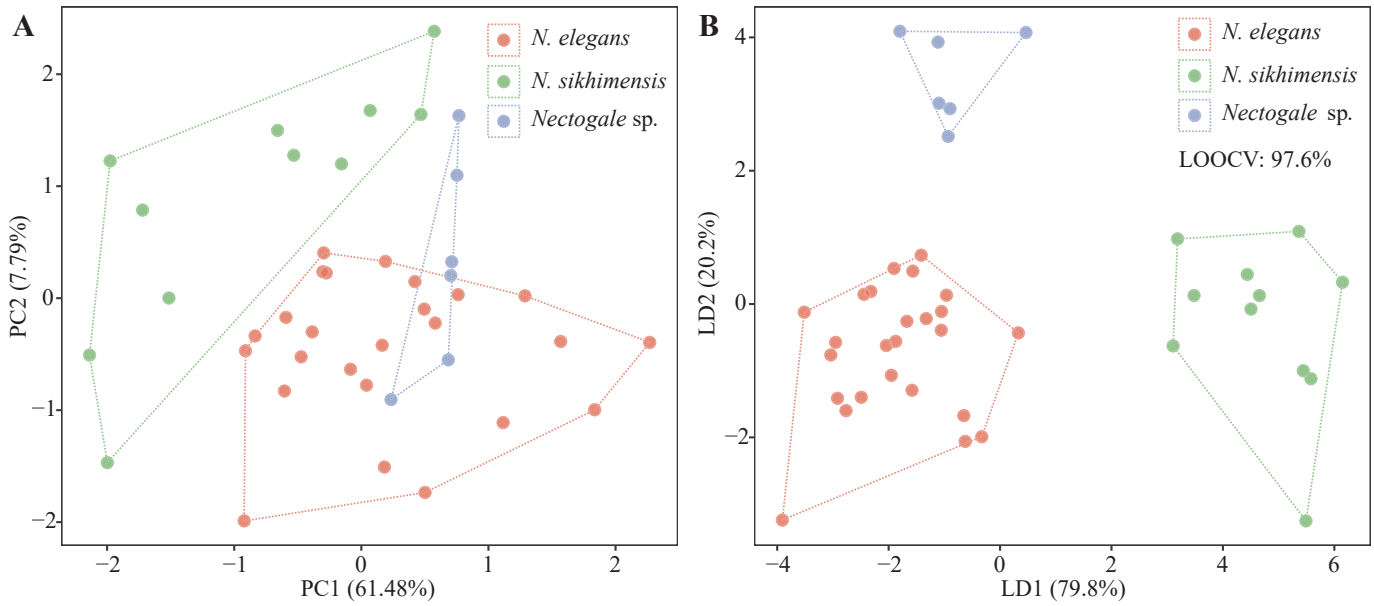


**Fig. 4.** Craniodental variation in *Nectogale*. (A) Principal component analysis (PCA) of all 22 $\log_{10}$-transformed craniodental measurements; axes show the first two principal components (PC1 and PC2) with the percentage of variance explained. (B) Linear discriminant analysis (LDA) based on the optimal subset of 12 variables (see Section 2.6); axes show the first two linear discriminants (LD1 and LD2) with the percentage of discriminant variance explained; LOOCV, leave-one-out cross-validation classification accuracy.

**Table 3**
Craniodental variation in *Nectogale* based on PCA. Variable loadings and variance explained (%) by the first two PCs are shown. Measurement abbreviations are defined in Supplementary Methods. Loadings > |0.7| are highlighted in bold.

| Variable | PC1 | PC2 |
|---|---|---|
| CIL | **0.927** | 0.241 |
| PIL | **0.907** | 0.162 |
| HCP | **0.900** | –0.078 |
| LLM | **0.885** | 0.112 |
| $M^2M^2$ | **0.878** | –0.108 |
| CB | **0.870** | 0.072 |
| LTR | **0.870** | 0.141 |
| MnL | **0.865** | –0.273 |
| UTL | **0.863** | 0.163 |
| MB | **0.858** | –0.209 |
| IOB | **0.845** | –0.187 |
| $P^4M^3$ | **0.842** | 0.039 |
| RL | **0.809** | –0.305 |
| PRL | **0.809** | 0.153 |
| PPL | **0.791** | 0.028 |
| HAC | **0.767** | 0.146 |
| CH | 0.698 | 0.145 |
| HCV | 0.675 | 0.038 |
| PPD | 0.671 | –0.489 |
| ZP | 0.496 | –0.428 |
| BMF | 0.349 | 0.545 |
| LLI | 0.170 | **0.717** |
| Variance explained | 61.485 | 7.795 |
| Eigenvalue | 13.527 | 1.715 |

interspersed lustrous white guard hairs. The tail base is dark gray, covered with short white bristles; caudal fimbriae are underdeveloped, and the tail-tip fimbriae are bright white, giving the tail an overall lighter appearance than in the other two species (Fig. 5C). Tail length overlaps with that of *N. elegans* and *N. sikhimensis* and slightly exceeds head–and–body length (TL/HBL ≈ 110%). The feet are light brownish gray to tan dorsally, with dark brown plantar surfaces and lustrous white fimbriae along the margins of the feet and digits.

The skull is similar in overall plan to that of congeners, with a flattened triangular outline. The braincase is laterally expanded, producing a wide and relatively flat cranial profile; it is flatter than that of *N. sikhimensis* but more inflated than that of *N. elegans*. The cranial surface is smooth, with well-developed sagittal and lambdoidal crests. The dental formula is consistent with the genus: 1.3.1.3/1.1.1.3 = 28. The principal (anterior) cusp of the upper incisor (I1) is well developed, whereas the posterior cusp is indistinct. Three upper unicuspids ($U^1$–$U^3$) are present, decreasing in both length and crown height from $U^1$ to $U^3$. All upper unicuspids have small but sharp cusps, twisted bases, and thin cingula. $P^4$ has a triangular outline, is smaller than $M^1$, and is separated from $U^3$ by a prominent diastema. The posterior edges of $P^4$ and $M^1$ are deeply and triangularly concave. The three upper molars decrease in size from $M^1$ to $M^3$, with $M^3$ approximately half the size of $M^2$ (Fig. 6C2).

The mandible is robust. The coronoid process is high, straight, and relatively broad, with a concave anterior surface and a wide, rounded apex; it is columnar overall, and its lateral surface bears a distinct bevel. The condyloid process is robust and triangular, with a slightly dorsally curved tip, forming an angle of approximately 120° with the coronoid process. The angular process is short and robust, shaped like a triangular pyramid; its apex does not exceed the condyloid process. The lower incisor ($I_1$) is slender and slightly upcurved. The lower unicuspid tooth ($U_1$) is the smallest, positioned above the base of the lower incisor, and its posterior margin is closely adjacent to the lower premolar ($P_4$). The primary cusp of the lower premolar is larger than the secondary cusp. All three lower molars are M−shaped with sharp cusps, decreasing in size from $M_1$ to $M_3$ (Fig. 6C4).

**Comparison.** Based on their typical webbed feet, characteristic tail morphology with caudal fimbriae, and overall external appearance, the Rikaze specimens are assigned to *Nectogale* rather than to other semi-aquatic shrew taxa such as *Chimarrogale*, *Neomys*, or *Sorex palustris*. Within the genus, *N. jilongensis* sp. nov. is morphologically similar to *N. elegans* and *N. sikhimensis*, but differs from both in several diagnostic traits. It is the largest species in the genus with a head–and–body length (HBL) of 105.50 ± 6.75 mm, a tail length (TL) of 115.67 ± 2.73 mm, a condyle-incisive length (CIL) of 26.02 ± 0.38 mm, and a mandibular length (MnL) of 12.29 ± 0.17 mm (Table 2), although measurements partially overlap with those of *N. elegans* and *N. sikhimensis*. Dorsal pelage is gray (lightest among the three species), whereas *N. elegans* is brownish gray and *N. sikhimensis* is grayish black; all three species have lustrous white guard hairs (Fig. 5A1, B1, C1). The tail of *N. jilongensis* sp.

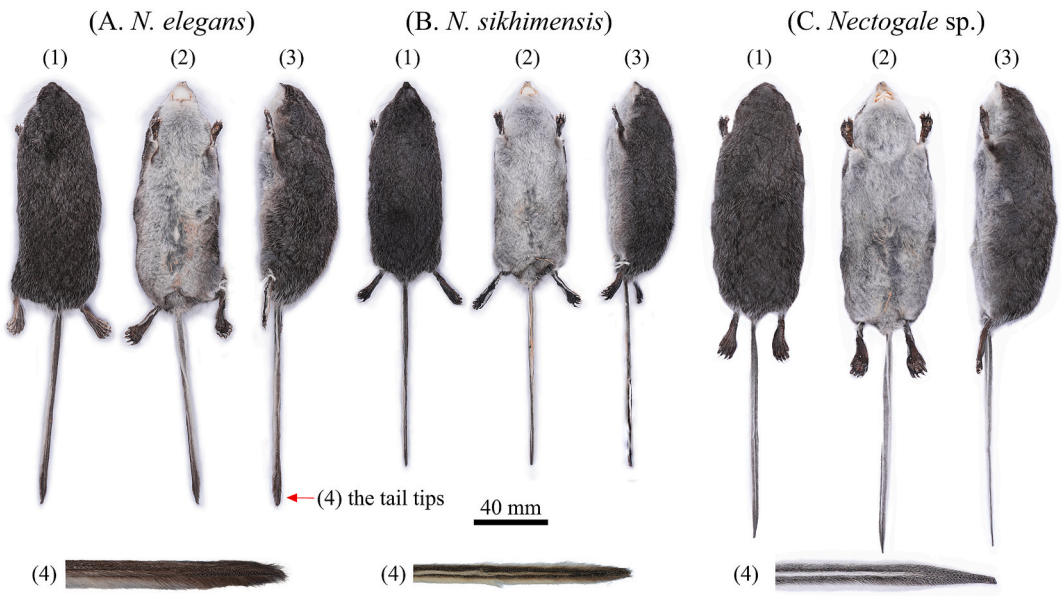


**Fig. 5.** Skins of (A) *N. elegans* (EG202311359), (B) *N. sikhimensis* (CN202303387), and (C) *Nectogale* sp. (JL202303775, holotype) in dorsal (1), ventral (2) and lateral (3) view; (4) enlarged views of the tail tips showing the caudal fimbriae. Scale bar: 40 mm (panels 1–3).

nov. bears short, bright white fimbriae that are sparse at the tip (Fig. 5C4), whereas *N. elegans* has well-developed caudal fimbriae with dense, long, reddish-brown fimbriae at the tail tip, and *N. sikhimensis* has sparse, short brown fimbriae at the tail tip (Fig. 5A4, B4).

Upper unicuspids of *N. jilongensis* sp. nov. have twisted bases, thin cingula, and small but sharp cusps; the length order is $U^1 > U^2 > U^3$ (Fig. 6C6). In *N. elegans* and *N. sikhimensis*, the upper unicuspids are of more typical form. In *N. elegans*, cingula are thicker, giving a more robust appearance; $U^1$ and $U^2$ are similar in size and $U^3$ is slightly smaller. In *N. sikhimensis*, the unicuspids are straighter with thinner cingula; $U^1$ and $U^2$ are similar and $U^3$ is slightly smaller. The coronoid process of *N. jilongensis* sp. nov. is wide, robust, and columnar, with a distinct oblique lateral surface and a more angular overall outline. In contrast, the coronoid processes of *N. elegans* and *N. sikhimensis* narrow gradually and appear more rounded, that of *N. elegans* being the slenderest and that of *N. sikhimensis* intermediate.

Distribution and habits

*Nectogale jilongensis* sp. nov. is currently known only from two counties, Jilong (type locality) and Dingjie, in Rikaze (Shigatse), southern Xizang, China, on the southern slope of the Himalayas west of Mount Kangchenjunga. It inhabits mountain streams at elevations of approximately 2350–3060 m (Table S1). It is semi-aquatic and an adept swimmer. Field observations indicate that it feeds on aquatic invertebrates and fish and that its activity pattern is cathemeral.

## 4. Discussion

### 4.1. Phylogeny and species delimitation

Interspecific relationships within *Nectogale* have historically been difficult to resolve. External morphology is highly conserved, sampling has been limited across much of the genus' range, and previous phylogenetic datasets lacked resolution. The genus was long treated as monotypic until Fan et al. (2022) revalidated *N. sikhimensis* as a species. Our recent field surveys recovered additional *Nectogale* populations from the central Himalayas that we hypothesized to represent an undescribed taxon.

Phylogenetic analyses of nuclear and mitochondrial genomes consistently recovered three strongly supported lineages within *Nectogale* with identical ingroup topologies (Fig. 2); discordance was confined to distantly related outgroups and did not affect relationships within *Nectogale*. The magnitude of mitochondrial divergence supports deep lineage separation: K2P distances for Cyt b (7.7–13.9%) were comparable to or exceeded the mean divergence reported between mammalian sister species (8.1%; Bradley and Baker, 2001). Population genomic summaries were concordant with these results, showing strong differentiation among the three lineages (pairwise FST = 0.558–0.816; $D_{XY}$ = 0.03–0.058; Table 1). BPP also supported the three-species model (PP = 1.0; Table S4). Genome-wide SNP clustering was likewise consistent with species-level structure: PCA and admixture recovered three distinct clusters corresponding to the three lineages (Fig. 3A, C).

Although external morphology is broadly conserved across *Nectogale*, the three lineages differed significantly in craniodental measurements and in qualitative cranial and pelage characters (Table 2; Figs. 4–6). This concordant evidence from phylogenomics, species delimitation, population structure, and cranial morphometrics supports recognition of *N. jilongensis* sp. nov. as a distinct species.

### 4.2. Evolutionary history and gene flow

Divergence-time estimates dated the split between *Nectogale* and *Chimarrogale* to the Middle Miocene (~13.98 Ma; 95% HPD = 11.03–16.89 Ma; Fig. 2D), with diversification within *Nectogale* beginning in the Late Miocene (~7.08 Ma) and the most recent split at ~ 4.49 Ma. The estimated stem age, which was constrained by a calibration prior derived from the broader fossil-calibrated framework of Yuan et al. (2024), was slightly older than the estimate of Fan et al. (2022) but consistent with that of He et al. (2021).

The genus subsequently split into two major branches in the Late Miocene (~7.08 Ma; 95% HPD = 4.77–9.51 Ma): *N. elegans*, now distributed in the Hengduan Mountains, and the (*N. sikhimensis* + *N. jilongensis* sp. nov.) branch, now distributed in the Himalayas. This split coincides temporally with major geological and climatic changes in the southeastern Qinghai-Xizang Plateau. First, intensified uplift during the Late Miocene likely reshaped topography and drainage networks and strengthened high-elevation barriers between adjacent watersheds (Cao et al., 2025; Ding et al., 2017; Han et al., 2026; Jiang et al., 2025; Lu et al., 2022; Mosbrugger et al., 2018; Shen et al., 2022; Shen et al., 2025). These landscape changes provide a plausible mechanism for fragmentation of ancestral populations and the

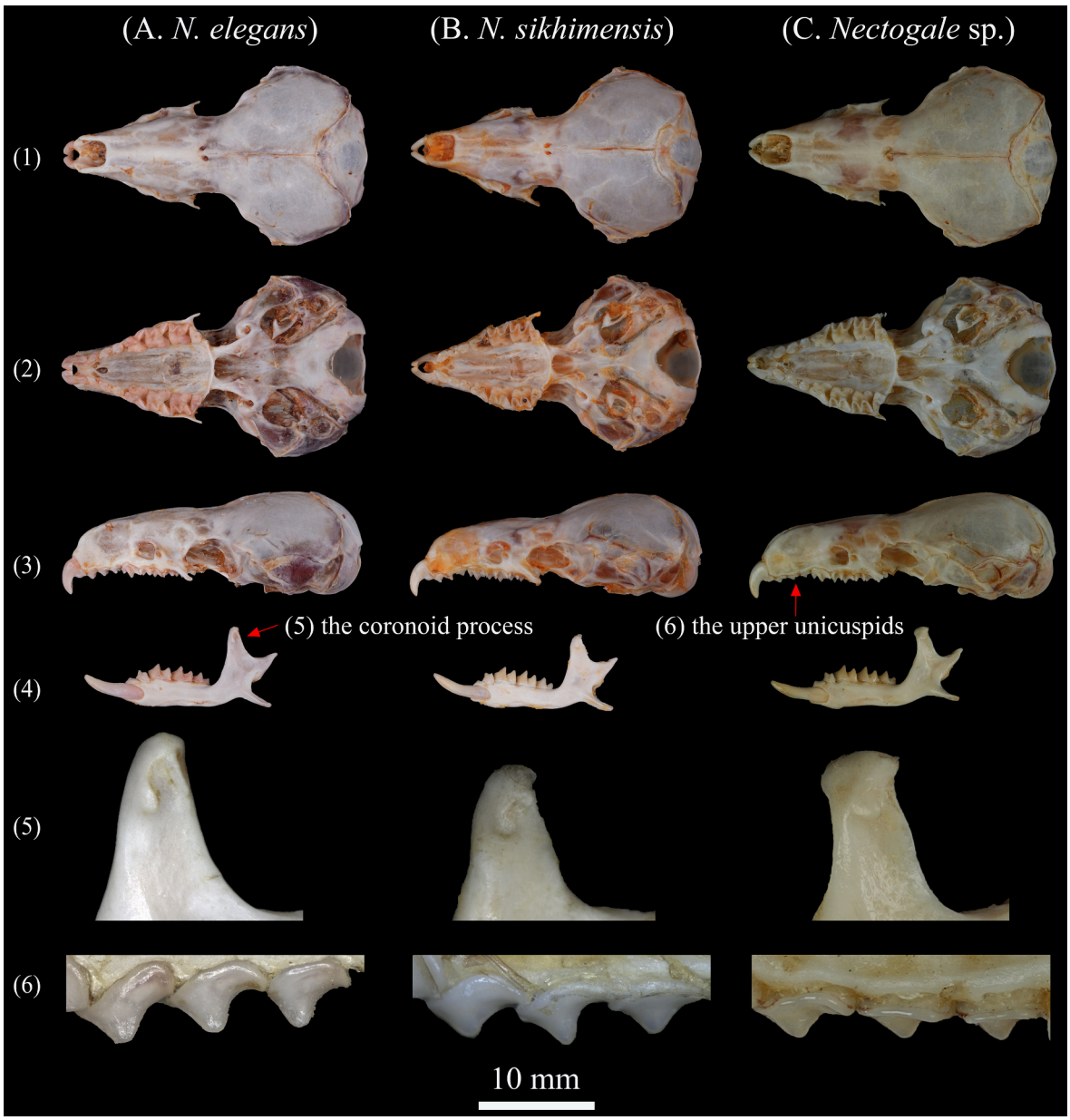


**Fig. 6.** Skulls and mandibles of (A) *N. elegans* (EG202311359), (B) *N. sikhimensis* (CN202303387), and (C) *Nectogale* sp. (JL202303775, holotype): (1) dorsal, (2) ventral and (3) lateral views of the skull; (4) lateral view of the mandible; (5) enlarged view of the coronoid process; (6) enlarged view of the upper unicuspids. Scale bar: 10 mm (panels 1–4); panels 5 and 6 are not to scale.

initiation of long-term isolation. Second, global cooling and an intensified winter monsoon in Southeast Asia during the Late Miocene may have further restricted suitable habitat and reinforced population isolation imposed by topographic barriers (Holbourn et al., 2018; Molnar et al., 1993; Molnar, 2005). The Himalayan branch subsequently diverged around the Early Pliocene (~4.49 Ma; 95% HPD = 2.64–6.48 Ma), giving rise to *N. sikhimensis* and *N. jilongensis* sp. nov., two species now distributed on opposite sides of the Mount Kangchenjunga massif. This divergence coincides with continued global cooling through the Pliocene (Qin et al., 2024; Westerhold et al., 2020; Zachos et al., 2001), and ongoing Himalayan uplift (Harrison et al., 1997), processes that increased regional topographic complexity and likely facilitated allopatric differentiation across major mountain barriers. Field surveys, specimen examination, and published records indicate clear distributional boundaries among the three species: the Boshula Ridge separates *N. elegans* from *N. sikhimensis*, and the Kangchenjunga massif separates *N. sikhimensis* from *N. jilongensis* sp. nov.

Quaternary glacial–interglacial cycles appear to have shaped both the regionally distinct demographic histories and the temporally restricted opportunities for interspecific gene flow. The PSMC reconstructions (Fig. 3E) showed that the three *Nectogale* species responded differently to environmental change. In southeastern Xizang, *N. sikhimensis* exhibited a brief population expansion during the Naynayxungla glaciation (~500–780 kya), which may relate to increased water vapor transport from the Bay of Bengal and development of relatively humid local microclimates (Shi, 1998). In contrast, *N. jilongensis* sp. nov. in the central Himalayas showed a modest expansion before the PG followed by long-term decline, consistent with drought stress associated with reduced moisture transport from the Arabian Sea (Shen and Jiao, 1989), whereas *N. elegans* in the Hengduan Mountains declined continuously, potentially driven by lower temperatures despite relatively humid conditions (Shi et al., 1995; Shi, 1998). During the Last Interglacial, warm and humid conditions may have allowed *N. sikhimensis* and *N. elegans* to recover slightly, while the

decline rate of *N. jilongensis* sp. nov. slowed, potentially creating short windows for interspecific contact. PSMC scales time by the estimated $N_0$ ($=\theta_0/4\mu$) of each genome (Li and Durbin, 2011). Because $\theta_0$ of *N. jilongensis* sp. nov. is lower than that of the other lineages, its PSMC curve spans a shorter time range and can be traced back only to before the PG. By the Last Glacial Maximum (~20 kya), intensified cooling and aridification likely caused pronounced population contractions in all three species (Li and Kang, 2006). PSMC has little resolution for the most recent tens of thousands of years (Li and Durbin, 2011), post-LGM dynamics cannot be inferred from these data.

Despite pronounced divergence among species, our genome-wide analyses revealed geographically localized introgression concentrated near species boundaries. Dsuite analyses detected a significant excess of allele sharing among lineages, with the strongest signals involving populations close to inferred contact zones, particularly among Himalayan populations of *N. sikhimensis*, *N. jilongensis* sp. nov., and western Yunnan population of *N. elegans*. ADMIXTURE provided complementary evidence: only three individuals, one from Bomi and two from Dingjie, showed mixed ancestry. Both localities lying at distributional margins, whereas all other individuals clustered with their respective species. These results suggest that gene flow in *Nectogale* is not widespread across species ranges but is concentrated in narrow zones of secondary contact.

The apparent discrepancy between genome-wide introgression signals inferred by ABBA–BABA statistics and the limited number of admixed individuals identified by admixture analyses likely reflects differences in temporal scale and methodological sensitivity. ABBA–BABA tests use genome-wide allele-sharing patterns and can detect accumulated, low-level introgression, potentially retaining signals from older events (Durand et al., 2011; Martin et al., 2015). By contrast, ADMIXTURE estimates individual ancestry proportions from current allele frequencies and is therefore more sensitive to recent or ongoing gene flow (Alexander et al., 2009). Taken together, the two approaches suggest that introgression in *Nectogale* has accumulated over evolutionary time at the population level, whereas recent admixture is restricted to a few individuals within narrow contact zones, consistent with expectations from hybrid zone theory: following secondary contact among differentiated lineages, hybridization is often localized, and the formation and persistence of hybrid zones reflect a balance between dispersal and selection against hybrids (Barton and Hewitt, 1985; Harrison, 1993; Abbott et al., 2013).

The alternation between introgression and isolation inferred from these patterns may reflect cyclical changes in landscape connectivity across the Himalayas during Quaternary glacial–interglacial cycles. Paleoclimate reconstructions indicate that climatic oscillations reshaped hydrological systems and drainage networks across the Qinghai-Xizang Plateau (Shi, 1998; Shen et al., 2023; Wang et al., 2023), thereby altering opportunities for dispersal in stream-dependent taxa. In this context, relatively warm and humid interglacial periods may have enhanced hydrological connectivity and facilitated secondary contact among previously isolated populations, whereas colder and drier glacial phases may have restricted dispersal, reinforced isolation, and allowed introgressed alleles to persist or become differentiated within lineages (He et al., 2016).

### 4.3. Biogeography and conservation

The Himalayas are a global biodiversity hotspot whose diversity remains underestimated (Shrestha et al., 2022). Recent descriptions of new small-mammal taxa from across the Himalayas (Chen et al., 2024a; Chen et al., 2021, 2023, 2024b, 2026; Liu et al., 2017, 2022; Pan et al., 2025; Wang et al., 2024; Wang et al., 2025; Zhu et al., 2025), together with the discovery of *N. jilongensis* sp. nov., indicate that cryptic diversity persists across multiple subregions.

The evolutionary history of *Nectogale* illustrates several biogeographic features typical of Himalayan montane taxa. The lineage differentiation and demographic history inferred here support the view that the Himalayas have acted both as climatic refugia and as engines of diversification (Lei et al., 2015; Favre et al., 2015). The mountain barriers and intervening river valleys of the southeastern and central Himalayas appear to mark both distributional boundaries and contact zones for *Nectogale*, emphasizing their importance as dynamic biogeographic interfaces. These deeply incised valleys may function not only as barriers that promote isolation, but also as intermittent corridors that permit communication among otherwise fragmented stream-associated populations (Sun et al., 2017). The strong genetic differentiation detected over short geographic distances further highlights the capacity of Himalayan landscapes to generate and maintain biodiversity within highly restricted habitats. More broadly, the coexistence of long-term lineage persistence and episodic gene flow suggests an evolutionary history shaped by repeated shifts between isolation and connectivity, rather than by continuous, unidirectional fragmentation. Similar dynamics may also characterize other mountain-stream taxa in the region (Chen et al., 2009; Chafin et al., 2021).

From a conservation perspective, the long-term decline and currently low effective population size of *N. jilongensis* sp. nov. inferred by PSMC may indicate limited demographic resilience and potential vulnerability to future environmental change, particularly climate warming and hydrological alteration (Spielman et al., 2004; Frankham, 2005). This pattern is consistent with the hypothesis that regional environmental heterogeneity has produced contrasting evolutionary trajectories across the Himalayas, ranging from recurrent expansion–contraction dynamics in humid eastern regions to more persistent contraction in central and western regions (He et al., 2016; Li et al., 2019). Because *N. jilongensis* sp. nov. is restricted to high-elevation stream habitats and has a small known range and low inferred $N_e$, targeted protection of headwater habitats and riparian corridors may be important for reducing its extinction risk.

More broadly, our results support subdividing the Himalayan region into finer biogeographic units for future biological studies (Liu et al., 2025), because coarse regional delineations may obscure small-scale patterns of endemism and evolutionary distinctiveness (Myers et al., 2000). The presence of geographically structured genetic clades associated with local climatic and ecological conditions suggests that fine-scale environmental heterogeneity has contributed to the maintenance of diversity. A more detailed understanding of the distinct histories of individual watersheds and mountain ranges will therefore be essential for elucidating the origins of Himalayan biodiversity and for guiding conservation planning in this globally important region.

## CRediT authorship contribution statement

**Xiaoxin Pei:** Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Yixian Li:** Writing – review & editing, Software, Resources, Methodology. **Kenneth Otieno Onditi:** Writing – review & editing, Supervision, Methodology, Data curation, Conceptualization. **Hongjiao Wang:** Supervision, Resources, Project administration. **Weihua Qin:** Resources, Investigation. **Ren Zhu:** Resources, Investigation. **Quan Li:** Resources, Investigation, Funding acquisition. **Wenyu Song:** Validation, Resources, Investigation. **Xueyou Li:** Resources, Investigation, Funding acquisition. **Zhongxu Zhu:** Resources, Project administration, Investigation. **Zhongzheng Chen:** Writing – review & editing, Supervision, Resources, Methodology, Investigation. **Xuelong Jiang:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Data curation.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ympev.2026.108749.

## Data availability

Data will be made available on request.

The whole genome resequencing data reported in this study have been submitted to the NCBI Sequence Read Archive (SRA) under BioProject PRJNA1425748. All other data necessary to reproduce the analyses reported in this study can be accessed through the Zenodo link: https://zenodo.org/records/20838826.

## References

Abbott, R., Albach, D., Ansell, S., Arntzen, J.W., Baird, S.J.E., Bierne, N., Boughman, J., Brelsford, A., Buerkle, C.A., Buggs, R., Butlin, R.K., Dieckmann, U., Eroukhmanoff, F., Grill, A., Cahan, S.H., Hermansen, J.S., Hewitt, G., Hudson, A.G., Jiggins, C., Jones, J., Keller, B., Marczewski, T., Mallet, J., Martinez-Rodriguez, P., Möst, M., Mullen, S., Nichols, R., Nolte, A.W., Parisod, C., Pfennig, K., Rice, A.M., Ritchie, M.G., Seifert, B., Smadja, C.M., Stelkens, R., Szymura, J.M., Väinölä, R., Wolf, J.B.W., Zinner, D., 2013. Hybridization and speciation. J. Evol. Biol. 26, 229–246. https://doi.org/10.1111/j.1420-9101.2012.02599.x.

Alexander, D.H., Novembre, J., Lange, K., 2009. Fast model-based estimation of ancestry in unrelated individuals. Genome Res. 19, 1655–1664. https://doi.org/10.1101/gr.094052.109.

Allen, G.M., 1938. The mammals of China and Mongolia. American Museum of Natural History, New York.

Alonge, M., Lebeigle, L., Kirsche, M., Jenike, K., Ou, S., Aganezov, S., Wang, X., Lippman, Z.B., Schatz, M.C., Soyk, S., 2022. Automated assembly scaffolding using RagTag elevates a new tomato system for high-throughput genome editing. Genome Biol. 23, 258. https://doi.org/10.1186/s13059-022-02823-7.

Barton, N.H., Hewitt, G.M., 1985. Analysis of hybrid zones. Annu. Rev. Ecol. Syst. 16, 113–148. https://doi.org/10.1146/annurev.es.16.110185.000553.

Behr, A.A., Liu, K.Z., Gracie, L.F., Nakka, P., Ramachandran, S., 2016. Pong: fast analysis and visualization of latent clusters in population genetic data. Bioinformatics 32, 2817–2823. https://doi.org/10.1093/bioinformatics/btw327.

Bradley, R.D., Baker, R.J., 2001. A test of the genetic species concept: Cytochrome-b sequences and mammals. J. Mammal. 82, 960–973. https://doi.org/10.1644/1545-1542(2001)082%3C0960:ATOTGS%3E2.0.CO;2.

Burgin, C.J., He, K., 2018. Family Soricidae (shrews). In: Wilson, D.E., Mittermeier, R.A. (Eds.), Handbook of the Mammals of the World, Vol. 8: Insectivores, Sloths and Colugos. Lynx Edicions, Barcelona, pp. 332–551.

Cao, G.L., Li, X.Q., Xiang, K.L., Erst, A.S., Jabbour, F., Ortiz, R.D.C., Yang, J., Wang, W., 2025. Formation of three great asian plateaus, climate change, and biodiversity. Trends Ecol. Evol. 40, 970–982. https://doi.org/10.1016/j.tree.2025.07.008.

Capella-Gutiérrez, S., Silla-Martínez, J.M., Gabaldón, T., 2009. TrimAl: a tool for automated alignment trimming in large-scale phylogenetic analyses. Bioinformatics 25, 1972–1973. https://doi.org/10.1093/bioinformatics/btp348.

Chafin, T.K., Douglas, M.R., Edds, D.R., Douglas, M.E., 2021. Parallel introgression, not recurrent emergence, explains apparent elevational ecotypes of polyploid Himalayan snowtrout. R. Soc. Open Sci. 8, 210727. https://doi.org/10.1098/rsos.210727.

Chen, S.F., 2023. Ultrafast one-pass FASTQ data preprocessing, quality control, and deduplication using fastp. iMeta 2, e107. https://doi.org/10.1002/imt2.107.

Chen, W., Bi, K., Fu, J.Z., 2009. Frequent mitochondrial gene introgression among high elevation Tibetan megophryid frogs revealed by conflicting gene genealogies. Mol. Ecol. 18, 2856–2876. https://doi.org/10.1111/j.1365-294X.2009.04258.x.

Chen, Z.Z., He, K., Huang, C., Wan, T., Lin, L.K., Liu, S.Y., Jiang, X.L., 2017. Integrative systematic analyses of the genus *Chodsigoa* (Mammalia: Eulipotyphla: Soricidae), with descriptions of new species. Zool. J. Linn. Soc. 180, 694–713. https://doi.org/10.1093/zoolinnean/zlw017.

Chen, Z.Z., He, S.W., Hu, W.H., Song, W.Y., Onditi, K.O., Li, X.Y., Jiang, X.L., 2021. Morphology and phylogeny of scalopine moles (eulipotyphla: talpidae: scalopini) from the eastern Himalayas, with descriptions of a new genus and species. Zool. J. Linn. Soc. 193, 432–444. https://doi.org/10.1093/zoolinnean/zlaa172.

Chen, Z.Z., Hu, J.X., He, K., Zhang, B.W., Zhang, Y., Chu, J., Zhao, K., Onditi, K.O., Jiang, X.L., 2023. Molecular and morphological evidence support a new species of Asiatic short-tailed shrew (Eulipotyphla: Soricidae). J. Mammal. 104, 1455–1467. https://doi.org/10.1093/jmammal/gyad087.

Chen, S.D., Yang, S.Y., Qing, J., Fan, R.H., Tang, Z.Y., Liao, R., Guo, K.J., Zhang, M.F., Wu, N.F., Liu, S.Y., 2024a. Two new species of the genus musk shrew (*Crocidura*, Insectivora: Soricidae) discovered in Xizang. Acta Theriol. Sin. 44, 529–550. https://doi.org/10.16829/j.slxb.150612 (in Chinese with English abstract).

Chen, Z.Z., Pei, X.X., Hu, J.X., Song, W.Y., Laxman, K., Li, Q., Jiang, X.L., 2024b. Multilocus phylogeny and morphological analyses illuminate overlooked diversity of *Soriculus* (Mammalia: Eulipotyphla: Soricidae), with descriptions of two new endemic species from the eastern Himalayas. Zool. J. Linn. Soc. 201, 534–548. https://doi.org/10.1093/zoolinnean/zlad131.

Chen, Z.Z., Shen, Y.C., Zhang, M., Pei, X.X., Xu, Y.F., Song, W.Y., Zhu, Z.X., Onditi, K.O., Jiang, X.L., 2026. Systematic revision and species diversification of the genus *Episoriculus* (mammalia: eulipotyphla: soricidae), with descriptions of two new species: Zool. Philos. Phenomenol. Res. https://doi.org/10.24272/j.issn.2095-8137.2026.291.

Danecek, P., Auton, A., Abecasis, G., Albers, C.A., Banks, E., DePristo, M.A., Handsaker, R.E., Lunter, G., Marth, G.T., Sherry, S.T., McVean, G., Durbin, R., 1000 Genomes Project Analysis Group, 2011. The variant call format and VCFtools. Bioinformatics 27, 2156–2158. https://doi.org/10.1093/bioinformatics/btr330.

Danecek, P., Bonfield, J.K., Liddle, J., Marshall, J., Ohan, V., Pollard, M.O., Whitwham, A., Keane, T., McCarthy, S.A., Davies, R.M., Li, H., 2021. Twelve years of SAMtools and BCFtools. GigaScience 10, giab008. https://doi.org/10.1093/gigascience/giab008.

De Winton, W.E., Styan, F.W., 1899. On chinese mammals, principally from western Sechuen. with notes on Chinese squirrels. Proc. Zool. Soc. Lond. 1899, 572–578.

Ding, L., Spicer, R.A., Yang, J., Xu, Q., Cai, F.L., Li, S., Lai, Q.Z., Wang, H.Q., Spicer, T.E. V., Yue, Y.H., Shukla, A., Srivastava, G., Khan, M.A., Bera, S., Mehrotra, R., 2017. Quantifying the rise of the Himalaya orogen and implications for the south Asian monsoon. Geology 45, 215–218. https://doi.org/10.1130/G38583.1.

Ding, W.N., Ree, R.H., Spicer, R.A., Xing, Y.W., 2020. Ancient orogenic and monsoon-driven assembly of the world's richest temperate alpine flora. Science 369, 578–581. https://doi.org/10.1126/science.abb4484.

Ding, J.Y., Hua, D.L., Wei, Y.G., Yao, L.M., Wang, N., 2025. Widespread gene flow from *Betula tianshanica* and *Betula pendula* to *Betula microphylla*: a rare species from northwest China. Bot. J. Linn. Soc. 207, 47–55. https://doi.org/10.1093/botlinnean/boae034.

Durand, E.Y., Patterson, N., Reich, D., Slatkin, M., 2011. Testing for ancient admixture between closely related populations. Mol. Biol. Evol. 28, 2239–2252. https://doi.org/10.1093/molbev/msr048.

Ellerman, J.R., Morrison-Scott, T.C.S., 1951. Checklist of Palaearctic and Indian mammals 1758 to 1946, 2nd ed. British Museum (Natural History), London.

Emms, D.M., Kelly, S., 2019. OrthoFinder: phylogenetic orthology inference for comparative genomics. Genome Biol. 20, 238. https://doi.org/10.1186/s13059-019-1832-y.

Fan, R.H., Tang, K.Y., Dou, L., Fu, C.K., Faiz, A.ul H., Wang, X.M., Wang, Y.F., Chen, S.D., Liu, S.Y., 2022. Molecular phylogeny and taxonomy of the genus *Nectogale* (Mammalia: Eulipotyphla: Soricidae). Ecol. Evol. 12, e9404. https://doi.org/10.1002/ece3.9404.

Favre, A., Päckert, M., Pauls, S.U., Jähnig, S.C., Uhl, D., Michalak, I., Muellner-Riehl, A. N., 2015. The role of the uplift of the Qinghai-Tibetan plateau for the evolution of Tibetan biotas. Biol. Rev. 90, 236–253. https://doi.org/10.1111/brv.12107.

Feng, Z.J., Cai, G.Q., Zheng, C.L., 1986. The Mammals of Xizang. Science Press, Beijing (in Chinese).

Frankham, R., 2005. Genetics and extinction. Biol. Conserv. 126, 131–140. https://doi.org/10.1016/j.biocon.2005.05.002.

Guindon, S., Dufayard, J.-F., Lefort, V., Anisimova, M., Hordijk, W., Gascuel, O., 2010. New algorithms and methods to estimate maximum-likelihood phylogenies: assessing the performance of PhyML 3.0. Syst. Biol. 59, 307–321. https://doi.org/10.1093/sysbio/syq010.

Han, X., Dai, J.G., Liu, B.R., Yu, Z.C., 2026. Exhumation history of the eastern Tethyan Himalaya: evidence from apatite (U Th)/He thermochronology. Tectonophysics 920, 231035. https://doi.org/10.1016/j.tecto.2025.231035.

Harrison, R.G., 1993. Hybrid zones and the evolutionary process. Oxford University Press, London.

Harrison, T.M., Ryerson, F.J., Le Fort, P., Yin, A., Lovera, O.M., Catlos, E.J., 1997. A late Miocene-Pliocene origin for the central Himalayan inverted metamorphism. Earth Planet. Sci. Lett. 146, E1–E7. https://doi.org/10.1016/S0012-821X(96)00215-4.

He, D.K., Chen, Y.F., Liu, C.L., Tao, J., Ding, C.Z., Chen, Y.J., 2016. Comparative phylogeography and evolutionary history of schizothoracine fishes in the Changtang Plateau and their implications for the lake level and Pleistocene climate fluctuations. Ecol. Evol. 6, 656–674. https://doi.org/10.1002/ece3.1890.

He, K., Eastman, T.G., Czolacz, H., Li, S.H., Shinohara, A., Kawada, S., Springer, M.S., Berenbrink, M., Campbell, K.L., 2021. Myoglobin primary structure reveals multiple convergent transitions to semi-aquatic life in the world's smallest mammalian divers. eLife 10, e66797. https://doi.org/10.7554/eLife.66797.

Hoang, D.T., Chernomor, O., von Haeseler, A., Minh, B.Q., Vinh, L.S., 2018. UFBoot2: improving the ultrafast bootstrap approximation. Mol. Biol. Evol. 35, 518–522. https://doi.org/10.1093/molbev/msx281.

Hoffmann, R.S., 1987. A review of the systematics and distribution of Chinese red-toothed shrews (Mammalia: Soricinae). Acta Theriol. Sin. 7, 100–139.

Holbourn, A.E., Kuhnt, W., Clemens, S.C., Kochhann, K.G.D., Jöhnck, J., Lübbers, J., Andersen, N., 2018. Late Miocene climate cooling and intensification of Southeast Asian winter monsoon. Nat. Commun. 9, 1584. https://doi.org/10.1038/s41467-018-03950-1.

Huang, N., Li, H., 2023. Compleasm: a faster and more accurate reimplementation of BUSCO. Bioinformatics 39, btad595. https://doi.org/10.1093/bioinformatics/btad595.

Hutterer, R., 2005a. Homology of unicuspids and tooth nomenclature in shrews. In: Merritt, J.F., et al. (Eds.), Special Publication of the International Society of Shrew Biologists No. 1. International Society of Shrew Biologists, New York, pp. 397–404.

Hutterer, R., 2005b. Order Soricomorpha. In: Wilson, D.E., Reeder, D.A. (Eds.), Mammal Species of the World: a Taxonomic and Geographic Reference. Johns Hopkins University Press, Baltimore, pp. 220–311.

Jiang, H.J., Fu, C.K., Tang, K.Y., Li, F.J., Faiz, A.U.H., Guo, K.J., Liu, S.Y., Chen, S.D., 2023. Molecular phylogenetics and diversity of the Himalayan shrew (*Soriculus nigrescens* Gray, 1842) (Eulipotyphla, Soricidae) in Southwest China. Zootaxa 5263, 61–78. https://doi.org/10.11646/zootaxa.5263.1.3.

Jiang, J.W., Wu, C., Li, J., Haproff, P., Wang, G.S., Shen, X.M., Wu, H., Xia, Y., 2025. Late Cenozoic tectonic evolution of the Tengchong and Baoshan blocks, southeastern Tibetan Plateau: insights from low-temperature thermochronology. Tectonophysics 912, 230866. https://doi.org/10.1016/j.tecto.2025.230866.

Jin, J.J., Yu, W.B., Yang, J.B., Song, Y., dePamphilis, C.W., Yi, T.S., Li, D.Z., 2020. GetOrganelle: a fast and versatile toolkit for accurate de novo assembly of organelle genomes. Genome Biol. 21, 241. https://doi.org/10.1186/s13059-020-02154-5.

Kalyaanamoorthy, S., Minh, B.Q., Wong, T.K.F., von Haeseler, A., Jermiin, L.S., 2017. ModelFinder: fast model selection for accurate phylogenetic estimates. Nat. Methods 14, 587–589. https://doi.org/10.1038/nmeth.4285.

Katoh, K., Standley, D.M., 2013. MAFFT multiple sequence alignment software version 7: improvements in performance and usability. Mol. Biol. Evol. 30, 772–780. https://doi.org/10.1093/molbev/mst010.

Korunes, K.L., Samuk, K., 2021. pixy: unbiased estimation of nucleotide diversity and divergence in the presence of missing data. Mol. Ecol. Resour. 21, 1359–1368. https://doi.org/10.1111/1755-0998.13326.

Kumar, S., Stecher, G., Li, M., Knyaz, C., Tamura, K., 2018. MEGA X: molecular evolutionary genetics analysis across computing platforms. Mol. Biol. Evol. 35, 1547–1549. https://doi.org/10.1093/molbev/msy096.

Lai, Y.J., Wen, J., Zhou, Z.K., Ge, S., Spicer, R.A., Chen, Z.D., Chen, Y.Y., 2025. Uplift history and biological evolution of the Himalaya (I). J. Syst. Evol. 63, 1–4. https://doi.org/10.1111/jse.13171.

Lei, F.M., Qu, Y.H., Song, G., Alström, P., Fjeldså, J., 2015. The potential drivers in forming avian biodiversity hotspots in the east himalaya mountains of southwest China. Integr. Zool. 10, 171–181. https://doi.org/10.1111/1749-4877.12121.

Li, H., 2013. Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM. https://doi.org/10.48550/arXiv.1303.3997.

Li, H., Durbin, R., 2011. Inference of human population history from individual whole–genome sequences. Nature 475, 493–496. https://doi.org/10.1038/nature10231.

Li, C.L., Kang, S.C., 2006. Review of studies in climate change over the Tibetan Plateau. Acta Geogr. Sin. 61, 327–335 in Chinese with English abstract.

Li, D.H., Liu, C.M., Luo, R.B., Sadakane, K., Lam, T.W., 2015. MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct de Bruijn graph. Bioinformatics 31, 1674–1676. https://doi.org/10.1093/bioinformatics/btv033.

Li, M.J., Xie, D.F., Xie, C., Deng, Y.Q., Zhong, Y., Yu, Y., He, X.J., 2019. A phytogeographic divide along the 500 mm isohyet in the Qinghai-Tibet Plateau: insights from the phylogeographic evidence of Chinese Alliums (Amaryllidaceae). Front. Plant Sci. 10, 149. https://doi.org/10.3389/fpls.2019.00149.

Li, Z.T., Fan, H.Z., Liao, Z.Y., Wang, Y.X., Wei, F.W., 2025. Global spatiotemporal patterns of demographic fluctuations in terrestrial vertebrates during the Late Pleistocene. Sci. Adv. 11, eadq3938. https://doi.org/10.1126/sciadv.adq3938.

Liu, S.Y., Jin, W., Liu, Y., Murphy, R.W., Lv, B., Hao, H.B., Liao, R., Sun, Z.Y., Tang, M.H., Chen, W.C., Fu, J.R., 2017. Taxonomic position of chinese voles of the tribe Arvicolini and the description of 2 new species from Xizang, China. J. Mammal. 98, 166–182. https://doi.org/10.1093/jmammal/gyw170.

Liu, S.Y., Zhou, C.R., Meng, G.L., Wan, T., Tang, M.K., Yang, C.T., Murphy, R.W., Fan, Z. X., Liu, Y., Zeng, T., Zhao, Y., Liu, S.L., 2022. Evolution and diversification of mountain voles (Rodentia: Cricetidae). Commun. Biol. 5, 1417. https://doi.org/10.1038/s42003-022-04371-z.

Liu, Y., Saqib, S., Lu, L.M., Lai, Y.J., Hu, H.H., Peng, D.X., Zaman, W., Zhao, L.N., Liu, B., Wang, Q., Lu, A.M., Hong, D.Y., Ye, J.F., Chen, Z.D., 2025. Phylogenetic regionalization of the Pan-Himalayan vascular flora. J. Syst. Evol. 63, 12–24. https://doi.org/10.1111/jse.13166.

Lu, Y., Cui, Y.C., Si, S.T., Kyaw, K.K., Myo, M.A., Zhu, W.L., Shao, L., 2022. Rapid Uplift period of Eastern Himalayan syntaxis: evidence from sedimentology in central Myanmar basin. Earth Sci.-J. China Univ. Geosci. 47, 2573. https://doi.org/10.3799/dqkx.2021.240.

Malinsky, M., Matschiner, M., Svardal, H., 2021. Dsuite – Fast D-statistics and related admixture evidence from VCF files. Mol. Ecol. Resour. 21, 584–595. https://doi.org/10.1111/1755-0998.13265.

Manish, K., Pandit, M.K., 2018. Geophysical upheavals and evolutionary diversification of plant species in the Himalaya. PeerJ 6, e5919.

Manni, M., Berkeley, M.R., Seppey, M., Simão, F.A., Zdobnov, E.M., 2021. BUSCO update: novel and streamlined workflows along with broader and deeper phylogenetic coverage for scoring of eukaryotic, prokaryotic, and viral genomes. Mol. Biol. Evol. 38, 4647–4654. https://doi.org/10.1093/molbev/msab199.

Martin, S.H., Davey, J.W., Jiggins, C.D., 2015. Evaluating the use of ABBA–BABA statistics to locate introgressed loci. Mol. Biol. Evol. 32, 244–257. https://doi.org/10.1093/molbev/msu269.

Meng, G.L., Li, Y.Y., Yang, C.T., Liu, S.L., 2019. MitoZ: a toolkit for animal mitochondrial genome assembly, annotation and visualization. Nucleic Acids Res. 47, e63.

Milne-Edwards, A., 1870. Note sur quelques mammifères du Tibet oriental. C. R. Acad. Sci. 70, 341–342.

Minh, B.Q., Schmidt, H.A., Chernomor, O., Schrempf, D., Woodhams, M.D., von Haeseler, A., Lanfear, R., 2020. IQ-TREE 2: new models and efficient methods for phylogenetic inference in the genomic era. Mol. Biol. Evol. 37, 1530–1534. https://doi.org/10.1093/molbev/msaa015.

Molnar, P., 2005. Mio-Pliocene growth of the Tibetan Plateau and evolution of East Asian climate. Palaeontol. Electron. 8, 1–23.

Molnar, P., England, P., Martinod, J., 1993. Mantle dynamics, uplift of the Tibetan Plateau, and the Indian Monsoon. Rev. Geophys. 31, 357–396. https://doi.org/10.1029/93RG02030.

Mosbrugger, V., Favre, A., Muellner-Riehl, A., Päckert, M., Mulch, A., 2018. Cenozoic evolution of geo-biodiversity in the Tibeto-Himalayan region, in: Mountains, Climate, and Biodiversity. John Wiley & Sons Ltd, Oxford, UK, pp. 429–449.

Myers, N., Mittermeier, R.A., Mittermeier, C.G., da Fonseca, G.A.B., Kent, J., 2000. Biodiversity hotspots for conservation priorities. Nature 403, 853–858. https://doi.org/10.1038/35002501.

Pacifici, M., Santini, L., Di Marco, M., Baisero, D., Francucci, L., Grottolo Marasini, G., Visconti, P., Rondinini, C., 2013. Generation length for mammals. Nat. Conserv. 5, 89–94. https://doi.org/10.3897/natureconservation.5.5734.

Pandit, M.K., Manish, K., Sen, S., Bawa, K.S., Knoll, A.H., Webb, A.A.G., Clift, P.D., Condamine, F.L., Hoorn, C., 2025. Evolutionary triggers of plant speciation in the Himalaya-Tibet-Hengduan biodiversity epicentre. J. Biogeogr. 52, e15152. https://doi.org/10.1111/jbi.15152.

Pan, Q.H., Wang, Y.X., Yan, K., 2007. Field Guide to the Mammals of China. China Forestry Publishing House, Beijing (in Chinese).

Pan, X., Wang, X.M., Murphy, R.W., Peng, B.Q., Zhang, Z.Y., Liao, R., Chen, S.D., Liu, S. Y., 2025. Two new species of *Ochotona* (Mammalia, Lagomorpha) from Xizang. China. Ecol. Evol. 15, e71898. https://doi.org/10.1002/ece3.71898.

Pei, X.X., Chen, Z.Z., Li, Q., Li, X.Y., Pu, C.Z., Luo, K., Luo, J., Pu, M.J., Wang, H.J., Khanal, L., Jiang, X.L., 2024. A new species of the genus *Soriculus* (Soricidae, Eulipotyphla, Mammalia) from Medog in the Eastern Himalaya. Zookeys 1195, 139–155. https://doi.org/10.3897/zookeys.1195.115699.

Pertea, G., Pertea, M., 2020. GFF Utilities: GffRead and GffCompare. F1000Research 9, ISCB Comm J-304. https://doi.org/10.12688/f1000research.23297.2.

Purcell, S., Neale, B., Todd-Brown, K., Thomas, L., Ferreira, M.A.R., Bender, D., Maller, J., Sklar, P., de Bakker, P.I.W., Daly, M.J., Sham, P.C., 2007. PLINK: a tool set for whole–genome association and population-based linkage analyses. Am. J. Hum. Genet. 81, 559–575. https://doi.org/10.1086/519795.

Qin, X.Y., Liu, J., Song, A., Gao, Y., Chen, J.M., Li, S.F., Su, T., 2024. The rediscovery of the Pliocene flora from Mt. Cho Oyu, central Himalaya and its characteristic. Quat. Sci. 44 (6), 1495–1508. https://doi.org/10.11928/j.issn.1001-7410.2024.06.03 in Chinese.

Rambaut, A., Drummond, A.J., Xie, D., Baele, G., Suchard, M.A., 2018. Posterior summarization in Bayesian phylogenetics using Tracer 1.7. Syst. Biol. 67, 901–904. https://doi.org/10.1093/sysbio/syy032.

Revelle, W., 2026. psych: Procedures for Psychological, Psychometric, and Personality Research. Northwestern University, Evanston, Illinois. R package version 2.6.1, https://CRAN.R-project.org/package=psych.

Sharma, B., Hope, A., 2026. From mountain birth to mammalian marvels: a review of the himalayas' role in mammalian biogeography. J. Biogeogr. 53, e70216. https://doi.org/10.1111/jbi.70216.

Shen, Y.P., Jiao, K.Q., 1989. Discussion on the pedogenesis and environment of paleosols on the moraine plateform in Nyalam. Xizang. J. Glaciol. Geocryol. 11 (3), 269–278 in Chinese with English abstract.

Shen, X.M., Braun, J., Yuan, X.P., 2022. Southeastern margin of the Tibetan Plateau stopped expanding in the late Miocene. Earth Planet. Sci. Lett. 583, 117446. https://doi.org/10.1016/j.epsl.2022.117446.

Shen, Q.J., Zhou, Y.J., Xu, Y.T., Lai, Y.S., Yan, X., Huang, X.M., Liu, X.B., Zhong, J.M., Zhu, S., Li, Z.J., Lai, Z.P., 2023. Late Quaternary River evolution and its response to climate changes in the upper Mekong River of the Qinghai–Tibetan Plateau. Geomorphology 442, 108920. https://doi.org/10.1016/j.geomorph.2023.108920.

Shen, X.W., Jing, L.Z., Shen, X.M., Cao, K., Leloup, P.H., Qin, K.X., Xing, Y.K., Li, J.Y., Zeng, X.Y., Wang, H., 2025. Late Miocene extension of the Dali fault system in southeast Tibet: coupled tectonic inversion and landscape evolution. GSA Bull. 137, 5074–5094. https://doi.org/10.1130/B38058.1.

Shi, Y.F., 1998. Evolution of the cryosphere in the Tibetan Plateau, China, and its relationship with the global change in the mid quaternary. J. Glaciol. Geocryol. 20 (03), 197–208 in Chinese.

Shi, Y.F., Zheng, B.X., Li, S.J., Ye, B.S., 1995. Studies on altitude and climatic environment in the Middle and East Parts of Tibetan Plateau during quaternary maximum glaciation. J. Glaciol. Geocryol. 17 (02), 97–112 in Chinese with English abstract.

Shrestha, L., Sarkar, M.S., Shrestha, K., Aung, P.S., Wen, X., Yongping, Y., Zhipang, H., Xuefei, Y., Shaoliang, Y., Chettri, N., 2022. Mammalian research, diversity and conservation in the Far Eastern Himalaya landscape: a review. Glob. Ecol. Conserv. 34, e02003. https://doi.org/10.1016/j.gecco.2022.e02003.

Shumate, A., Salzberg, S.L., 2021. Liftoff: accurate mapping of gene annotations. Bioinformatics 37, 1639–1643. https://doi.org/10.1093/bioinformatics/btaa1016.

Sikes, R.S., Animal Care and Use Committee of the American Society of Mammalogists, 2016. 2016 Guidelines of the American Society of Mammalogists for the use of wild

mammals in research and education. J. Mammal. 97, 663–688. https://doi.org/10.1093/jmammal/gyw078.
Smith, A.T., Xie, Y., 2020. Mammals of China. Lynx Edicions, Barcelona.
Spielman, D., Brook, B.W., Frankham, R., 2004. Most species are not driven to extinction before genetic factors impact them. PNAS 101, 15261–15264. https://doi.org/10.1073/pnas.0403809101.
Sun, H., Zhang, J.W., Deng, T., Boufford, D.E., 2017. Origins and evolution of plant diversity in the Hengduan Mountains. China. Plant Divers. 39, 161–166. https://doi.org/10.1016/j.pld.2017.09.004.
Tonzo, V., Papadopoulou, A., Ortego, J., 2019. Genomic data reveal deep genetic structure but no support for current taxonomic designation in a grasshopper species complex. Mol. Ecol. 28, 3869–3886. https://doi.org/10.1111/mec.15189.
Venables, W.N., Ripley, B.D., 2002. Modern applied statistics with S, Statistics and Computing. Springer, New York. https://doi.org/10.1007/978-0-387-21706-2.
Walker, E.P., 1975. Mammals of the World. Johns Hopkins University Press, Baltimore.
Wang, Y.X., 2003. A Complete Checklist of Mammal Species and Subspecies in China: A Taxonomic and Geographic Reference. China Forestry Publishing House, Beijing (in Chinese).
Wang, P., Wang, H.Y., Hu, G., Ge, Y.K., Liu, T., Xu, B., 2023. Reconstructing post-mid Pleistocene glacial-dammed paleolakes in the Tsangpo Gorge, southeastern Tibetan Plateau. Quat. Sci. Rev. 314, 108228. https://doi.org/10.1016/j.quascirev.2023.108228.
Wang, S.Y., Li, Y.X., Li, Q., Song, W.Y., Wang, H.J., He, S.W., Onditi, K.O., Khanal, L., Li, X.Y., Chen, Z.Z., Jiang, X.L., 2024. A new species of mountain vole (Rodentia, Cricetidae, Neodon) from south Xizang, China. Zool. Res.: Divers Conserv. 1, 282–289. https://doi.org/10.24272/j.issn.2097-3772.2024.011.
Wang, X.M., Pan, X., Liu, Y.X., Murphy, R.W., Peng, B.Q., Duan, C., Liao, R., Wang, X., Liu, S.Y., 2025. Three new species of *Neodon* (Rodentia, Cricetidae) from Sichuan and Xizang. China. Zoosyst. Evol. 101, 999–1012. https://doi.org/10.3897/zse.101.140898.
Westerhold, T., Marwan, N., Drury, A.J., Liebrand, D., Agnini, C., Anagnostou, E., Barnet, J.S.K., Bohaty, S.M., De Vleeschouwer, D., Florindo, F., Frederichs, T., Hodell, D.A., Holbourn, A.E., Kroon, D., Lauretano, V., Littler, K., Lourens, L.J., Lyle, M., Pälike, H., Röhl, U., Tian, J., Wilkens, R.H., Wilson, P.A., Zachos, J.C., 2020. An astronomically dated record of Earth's climate and its predictability over the last 66 million years. Science 369, 1383–1387. https://doi.org/10.1126/science.aba6853.
Wilson, D.E., Mittermeier, R.A., 2018. Handbook of the Mammals of the World. Insectivores, Sloths and Colugos Vol. 8.
Xing, Y.W., Ree, R.H., 2017. Uplift-driven diversification in the Hengduan Mountains, a temperate biodiversity hotspot. PNAS 114, E3444–E3451. https://doi.org/10.1073/pnas.1616063114.
Xu, W., Dong, W.J., Fu, T.T., Gao, W., Lu, C.Q., Yan, F., Wu, Y.H., Jiang, K., Jin, J.Q., Chen, H.M., Zhang, Y.P., Hillis, D.M., Che, J., 2021. Herpetological phylogeographic analyses support a Miocene focal point of Himalayan uplift and biological diversification. Natl. Sci. Rev. 8, nwaa263. https://doi.org/10.1093/nsr/nwaa263.
Yang, Z.H., 2007. PAML 4: phylogenetic analysis by maximum likelihood. Mol. Biol. Evol. 24, 1586–1591. https://doi.org/10.1093/molbev/msm088.
Yang, Z.H., 2015. The BPP program for species tree estimation and species delimitation. Curr. Zool. 61, 854–865. https://doi.org/10.1093/czoolo/61.5.854.
Yang, Z.H., Rannala, B., 2010. Bayesian species delimitation using multilocus sequence data. PNAS 107, 9264–9269. https://doi.org/10.1073/pnas.0913022107.
Yang, Z.H., Rannala, B., 2014. Unguided species delimitation using DNA sequence data from multiple loci. Mol. Biol. Evol. 31, 3125–3135. https://doi.org/10.1093/molbev/msu279.
Yuan, H.B., Dickson, E.D., Martinez, Q., Arnold, P., Asher, R.J., 2024. The origin and evolution of shrews (Soricidae, Mammalia). Proc. R. Soc. B 291, 20241856. https://doi.org/10.1098/rspb.2024.1856.
Zachos, J., Pagani, M., Sloan, L., Thomas, E., Billups, K., 2001. Trends, rhythms, and aberrations in global climate 65 ma to present. Science 292, 686–693. https://doi.org/10.1126/science.1059412.
Zhang, R.Z., 1999. Distribution of Mammalian Species in China. China Forestry Publishing House, Beijing (in Chinese).
Zhang, D., Gao, F.L., Jakovlić, I., Zou, H., Zhang, J., Li, W.X., Wang, G.T., 2020. PhyloSuite: an integrated and scalable desktop platform for streamlined molecular sequence data management and evolutionary phylogenetics studies. Mol. Ecol. Resour. 20, 348–355. https://doi.org/10.1111/1755-0998.13096.
Zhang, C., Nielsen, R., Mirarab, S., 2025. CASTER: Direct species tree inference from whole–genome alignments. Science 387, eadk9688. https://doi.org/10.1126/science.adk9688.
Zhu, Z.X., Li, Q., Song, W.Y., Li, X.Y., Lissovsky, A., Wang, M.Y., Pei, X.X., Luo, K., Luo, J., Pu, M.J., Pu, C.Z., Wang, H.J., Liu, Z., Chen, Z.Z., Jiang, X.L., 2025. Discovery of an ancient Himalayan birch mouse lineage illuminates the evolution of the family Sicistidae (Rodentia: Dipodoidea), with descriptions of a new genus and two new species. Zool. Res. 46, 921–938. https://doi.org/10.24272/j.issn.2095-8137.2025.013.